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(54) OPTICAL SWITCH

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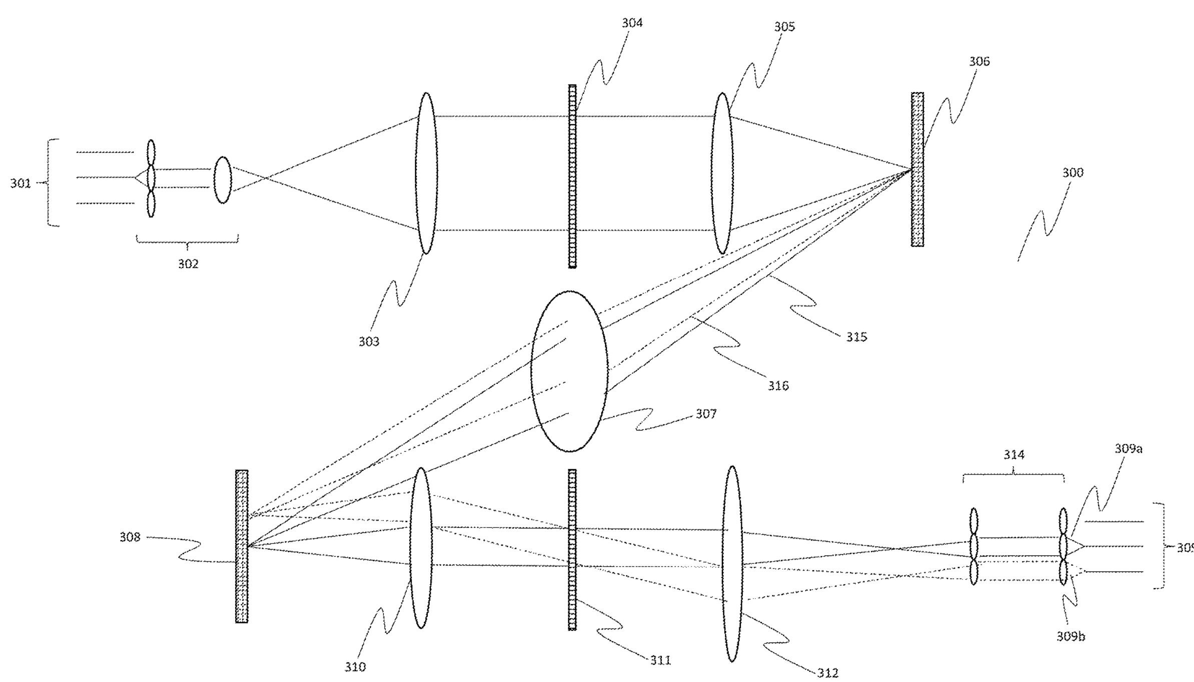
(57) **ABSTRACT**

An optical switch comprising: a set of first input ports. each first input port configured to transport an optical signal having at least one component frequency channel: a set of first output ports. each first output port configured to transport an optical signal having at least one component frequency channel: a first programmable deflection plane configured to deflect beams incident on it to form a first deflected array of beams: a second programmable deflection plane configured to deflect beams incident on it to form a second deflected array of beams: a first demultiplexer configured to separate light from the set of first input ports into its component frequency channels to form the beams incident on the first programmable deflection plane: a first multiplexer configured to combine the second deflected array of beams from the second programmable deflection plane into combined signals incident on the set of first output ports: and a beam steering optical element group configured to transfer the first deflected array of beams from the first programmable deflection plane to the beams incident on the second programmable deflection plane. The incidence normal of the first programmable



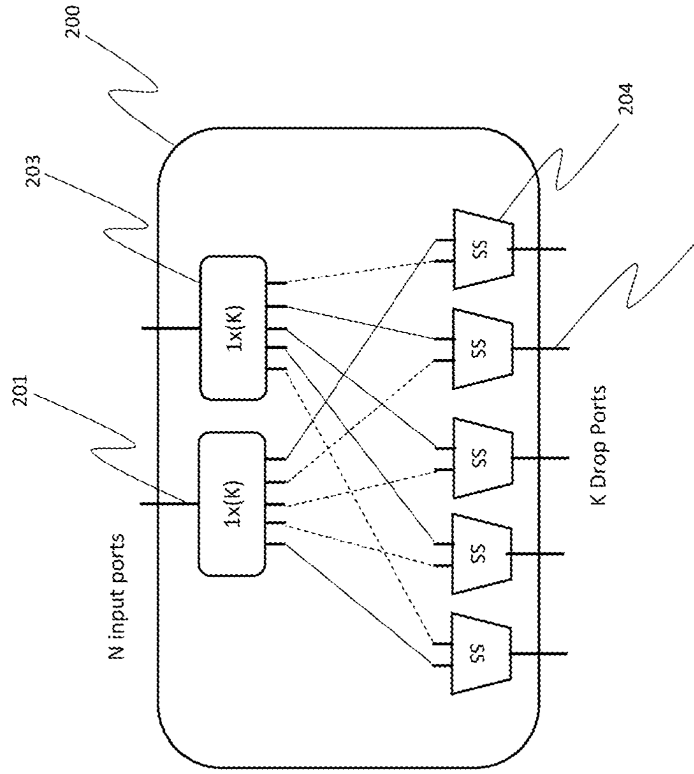


Figure 2

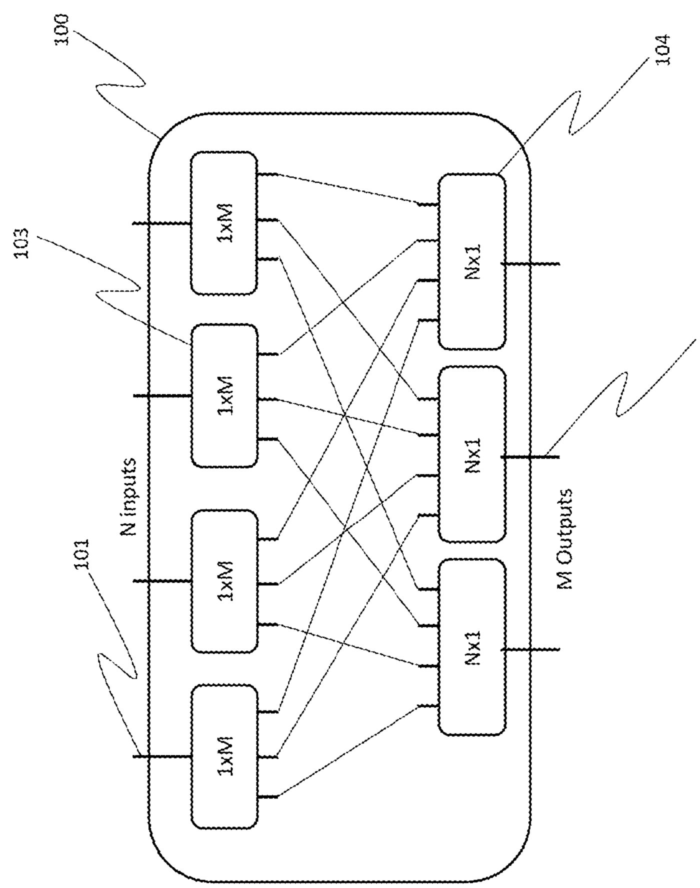


Figure 1

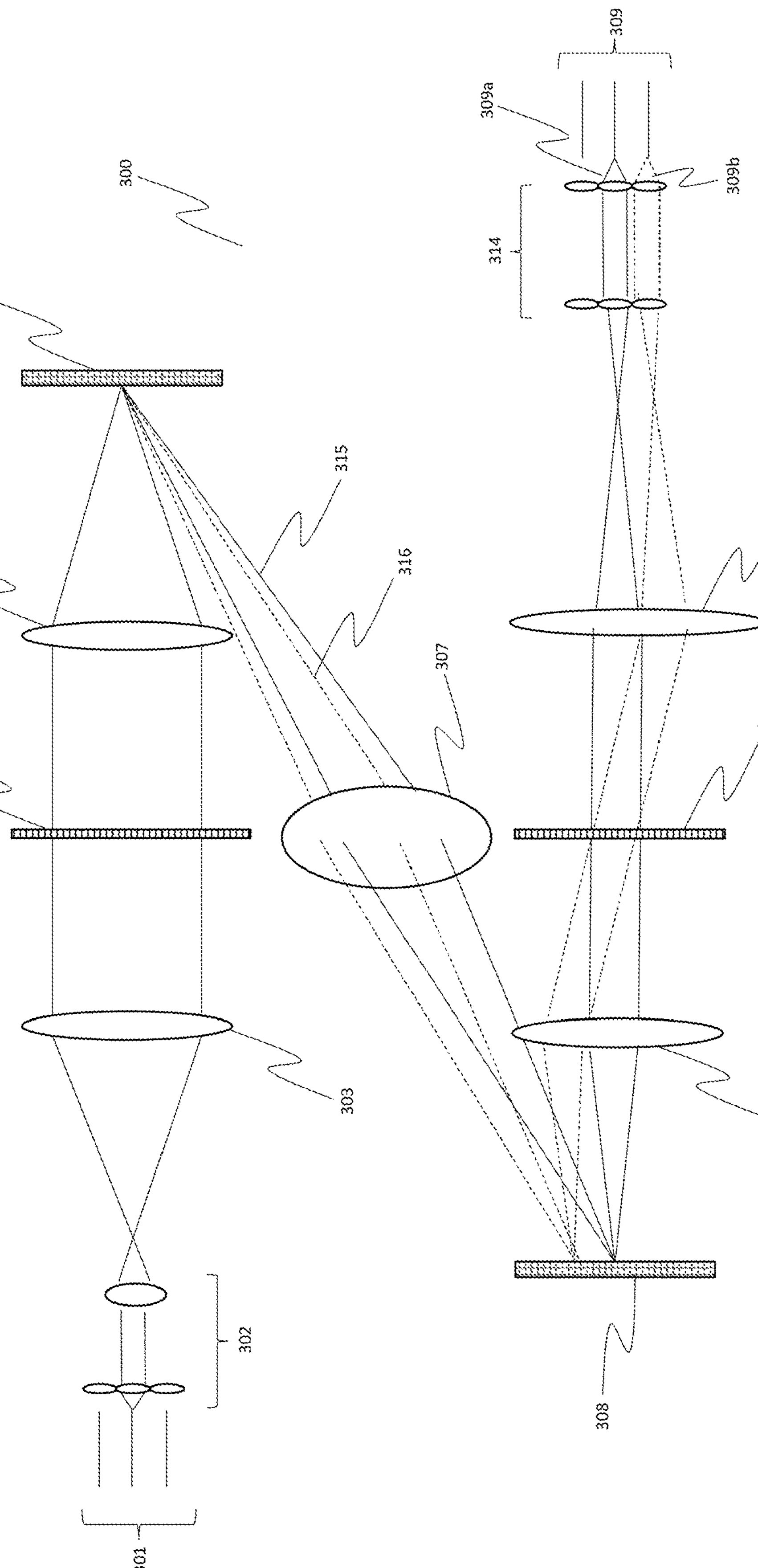

300
301
302
303
304
305
306
307
308
309
309a
309b
310
311
312
314
315
316

Figure 3

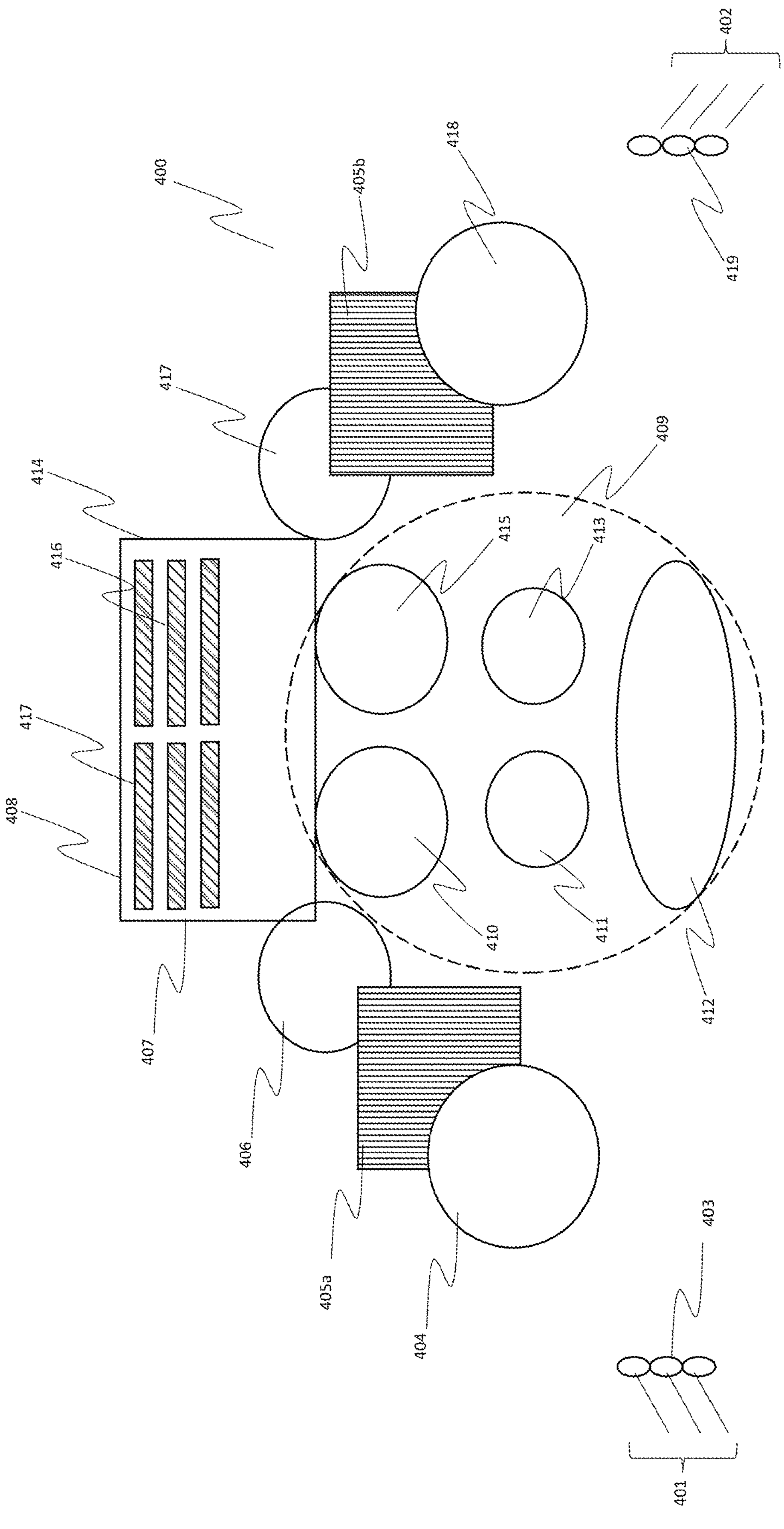

400
401
402
403
404
405a
405b
406
407
408
409
410
411
412
413
414
415
416
417
417
418
419

Figure 4

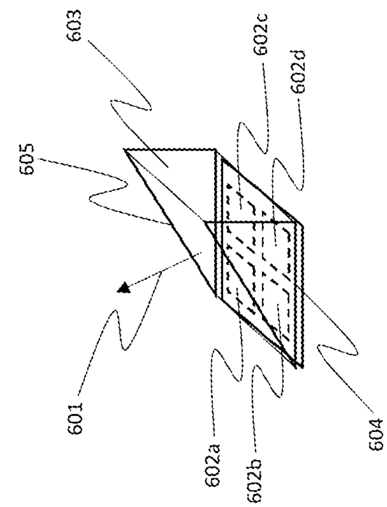


Figure 6

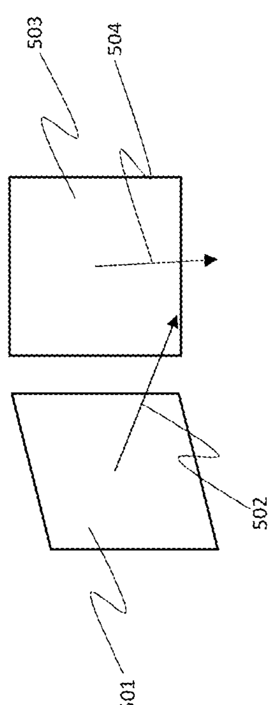


Figure 5

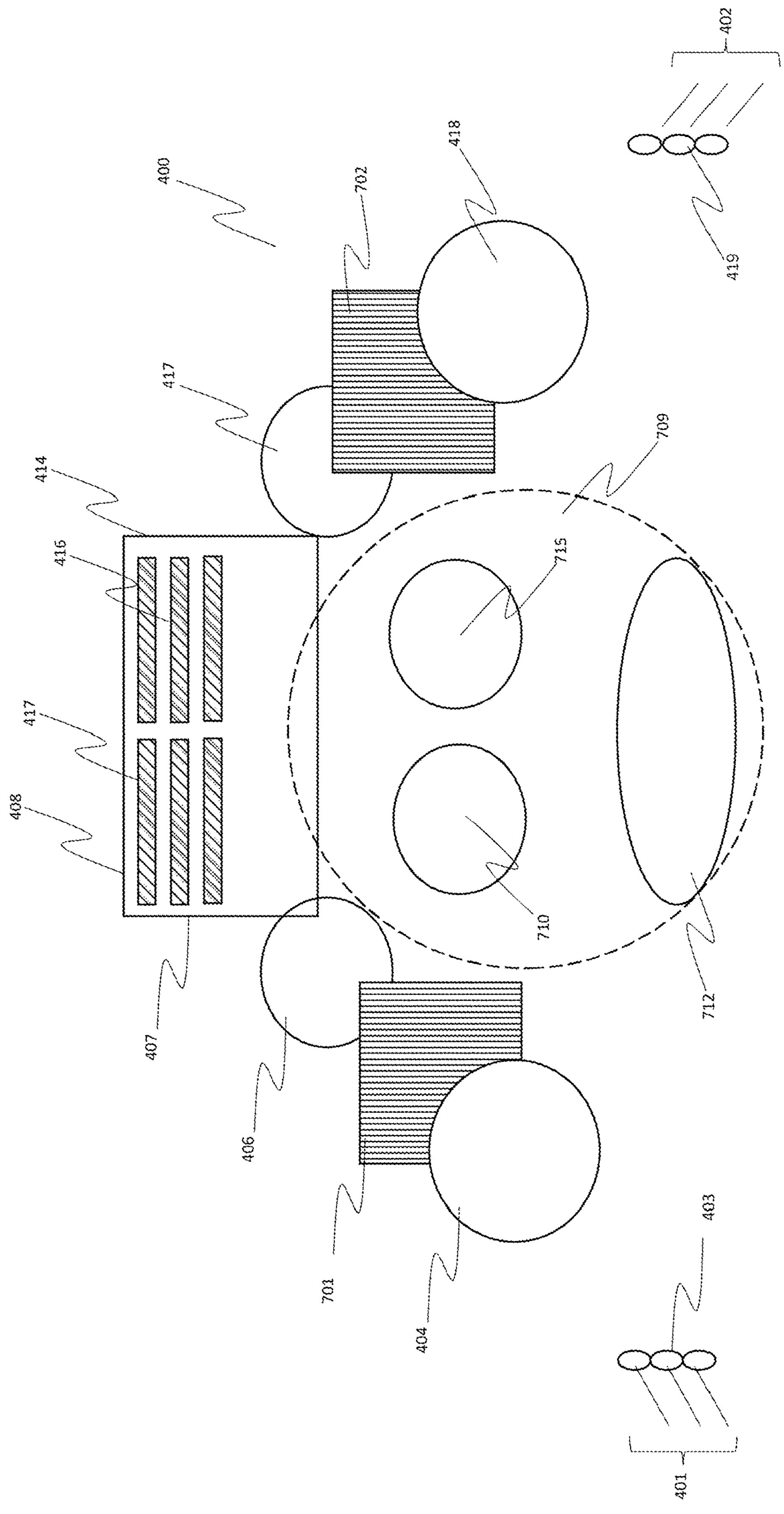

400
402
418
702
419
417
414
709
416
715
417
408
710
712
407
406
403
701
404
401

Figure 7

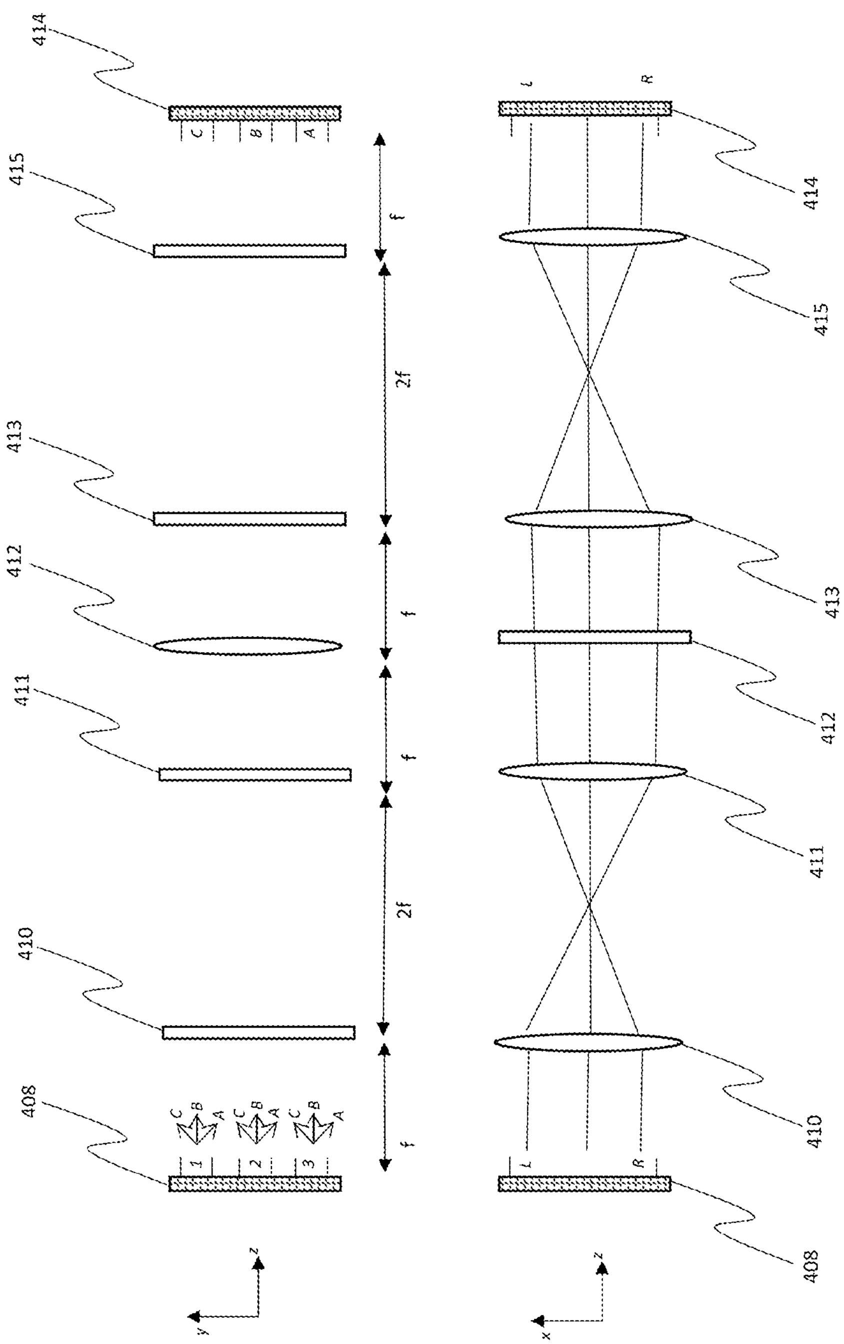


Figure 8

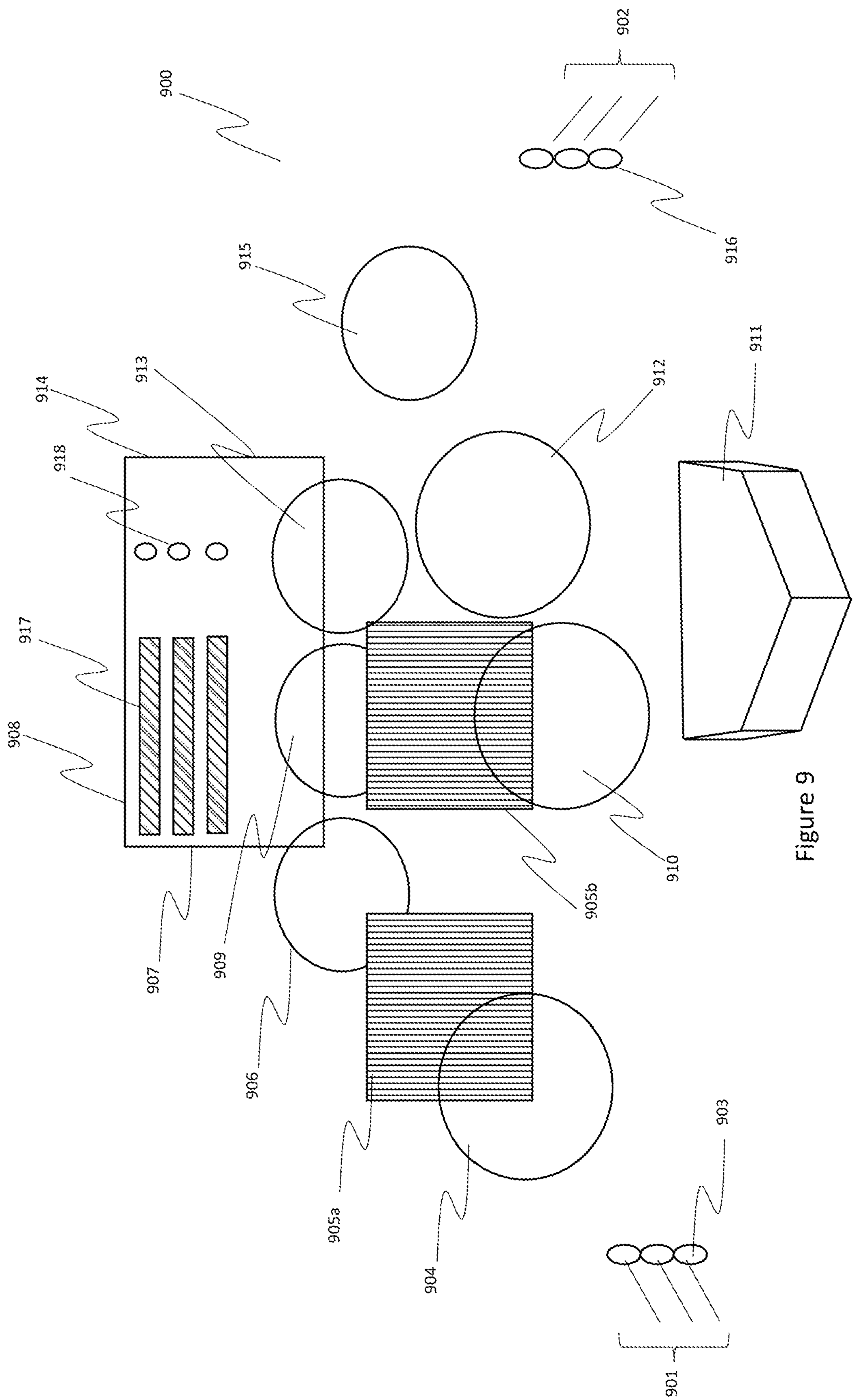

900
901
902
903
904
905a
905b
906
907
908
909
910
911
912
913
914
915
916
917
918

Figure 9

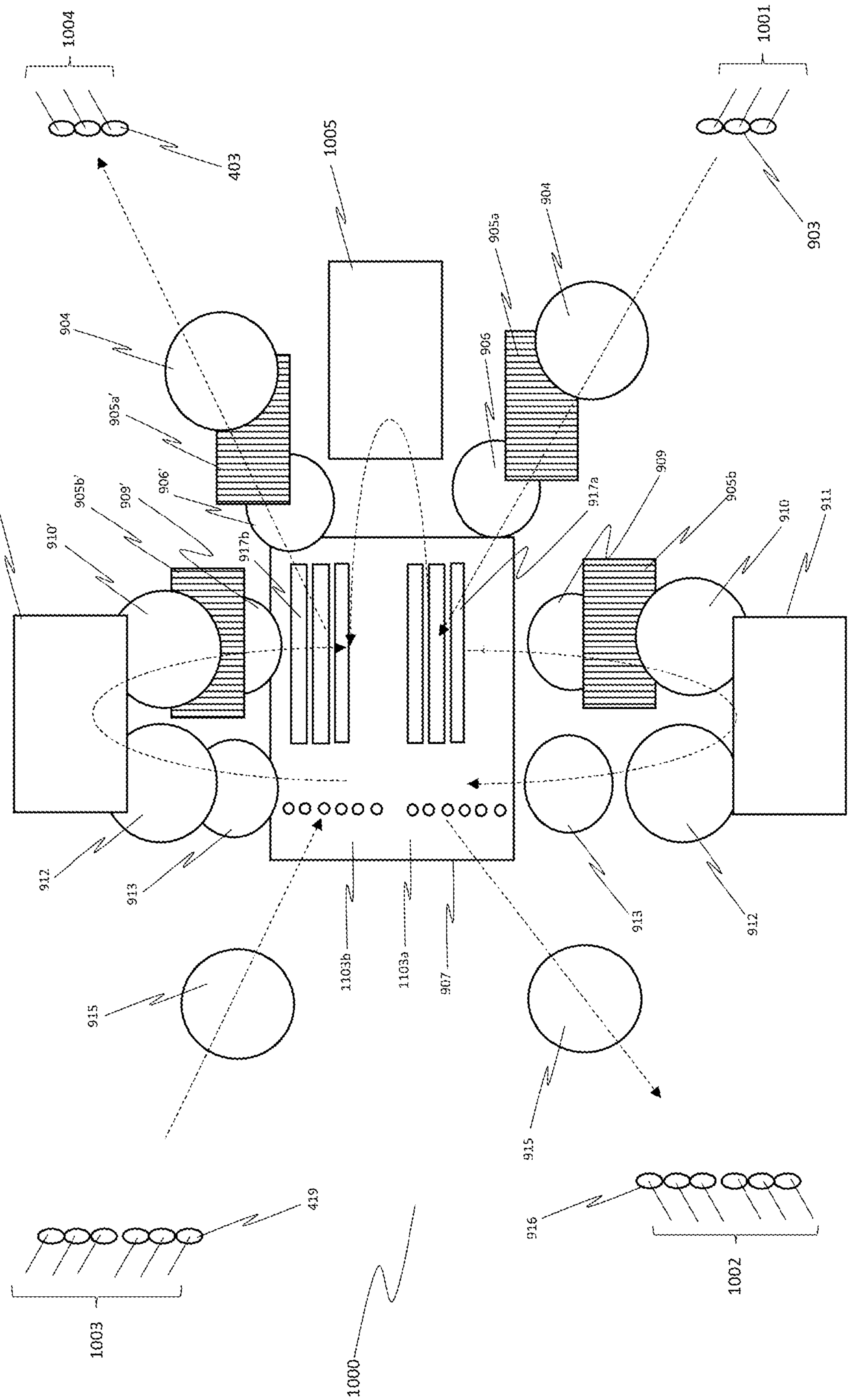

1004
1001
403
903
1005
904
905a
904
906
905a'
909
917a
905b
911'
905b'
909'
906'
910'
917b
910
911
912
913
913
912
1103b
1103a
907
915
915
419
916
1003
1002
1000

Figure 10

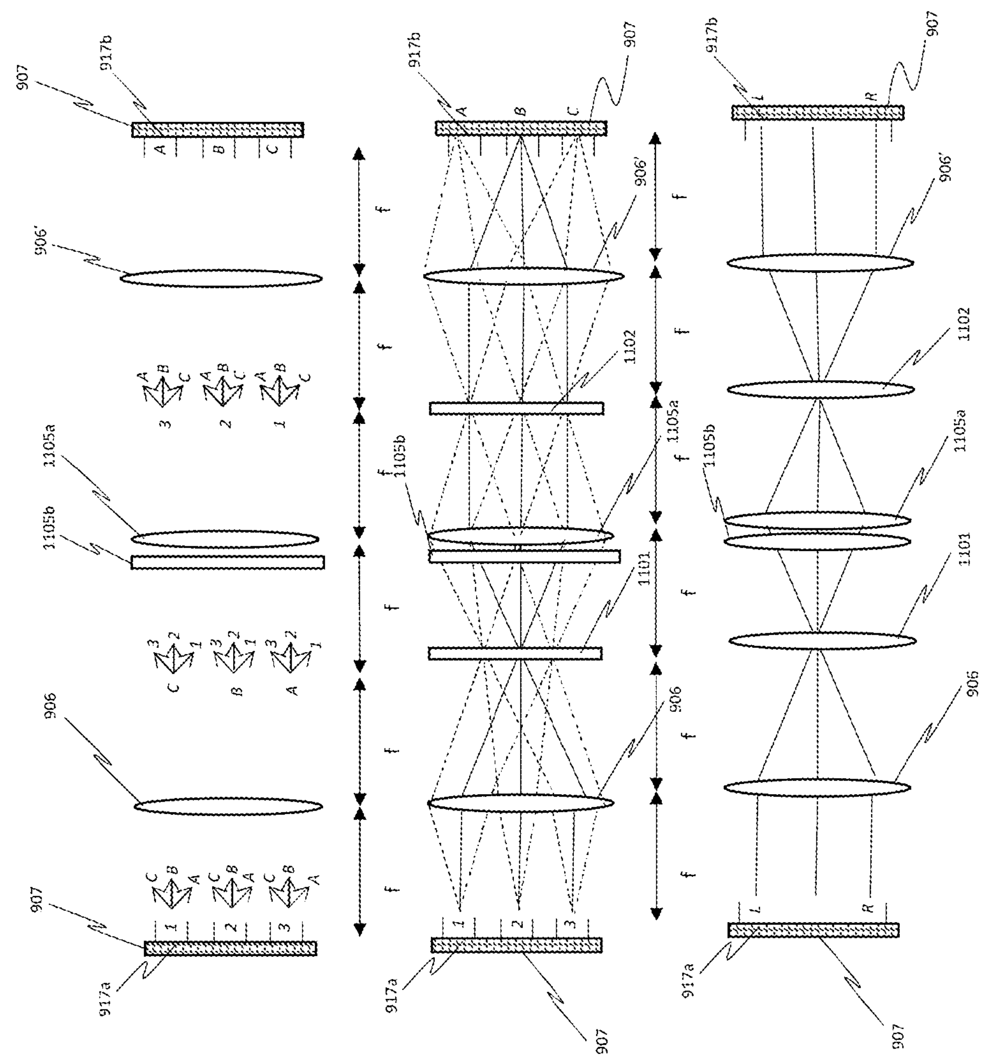


Figure 11

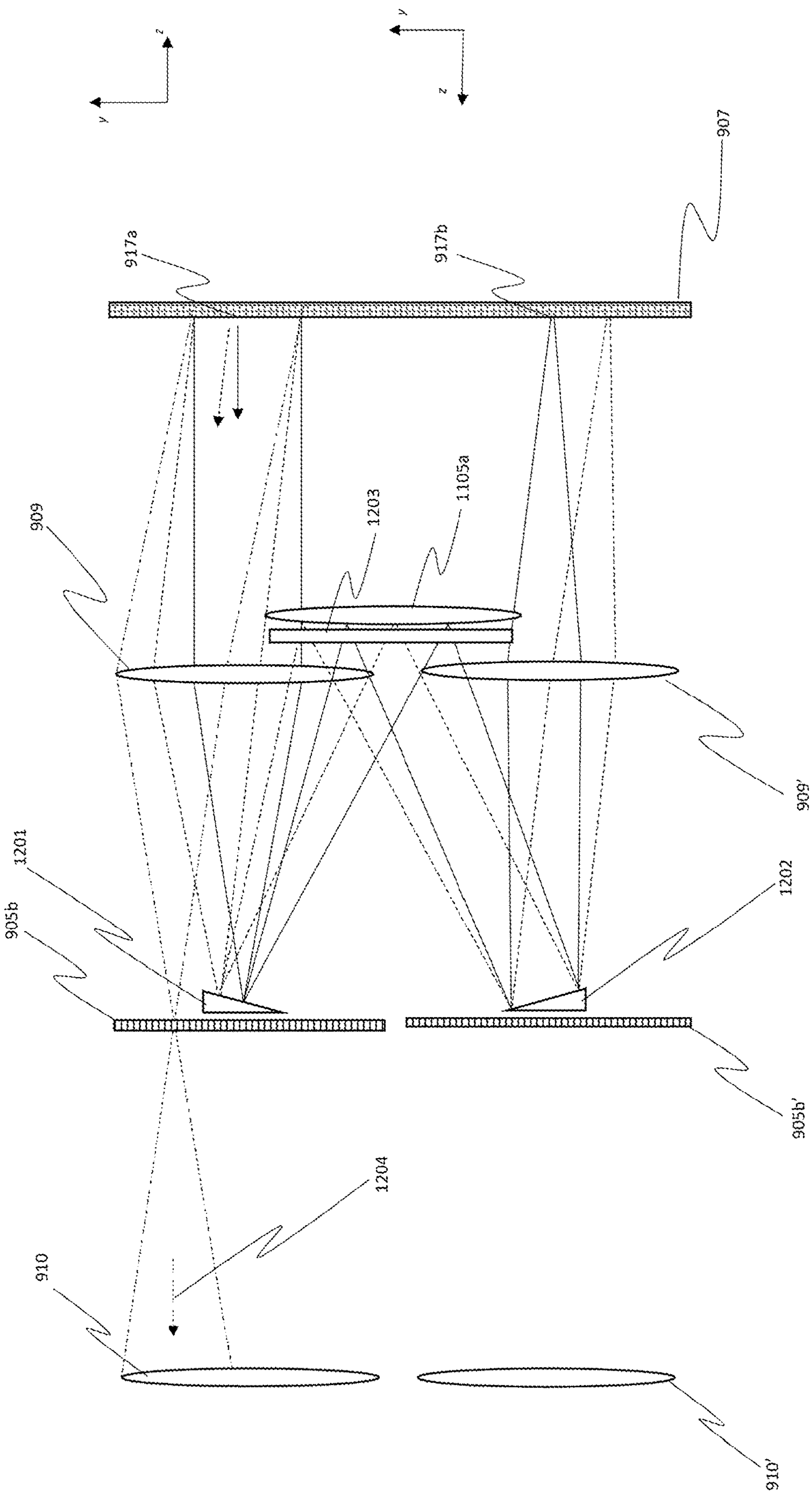
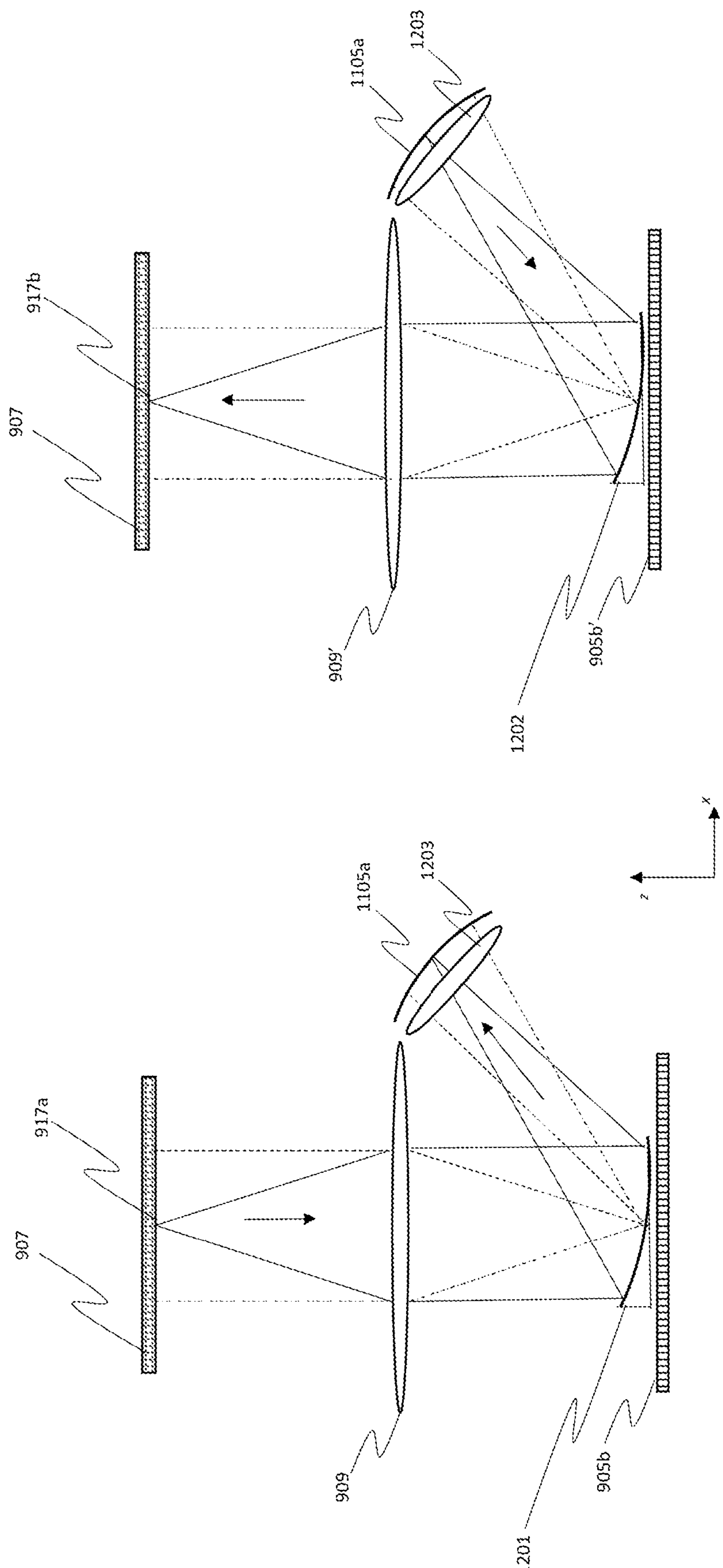

z
y
y
z
907
917a
917b
909
1203
1105a
909'
1201
905b
1202
905b'
1204
910
910'

Figure 12a



907
917b
909'
1105a
1203
905b'
1202
x
z
907
917a
909
1105a
1203
905b
1201

Figure 12b

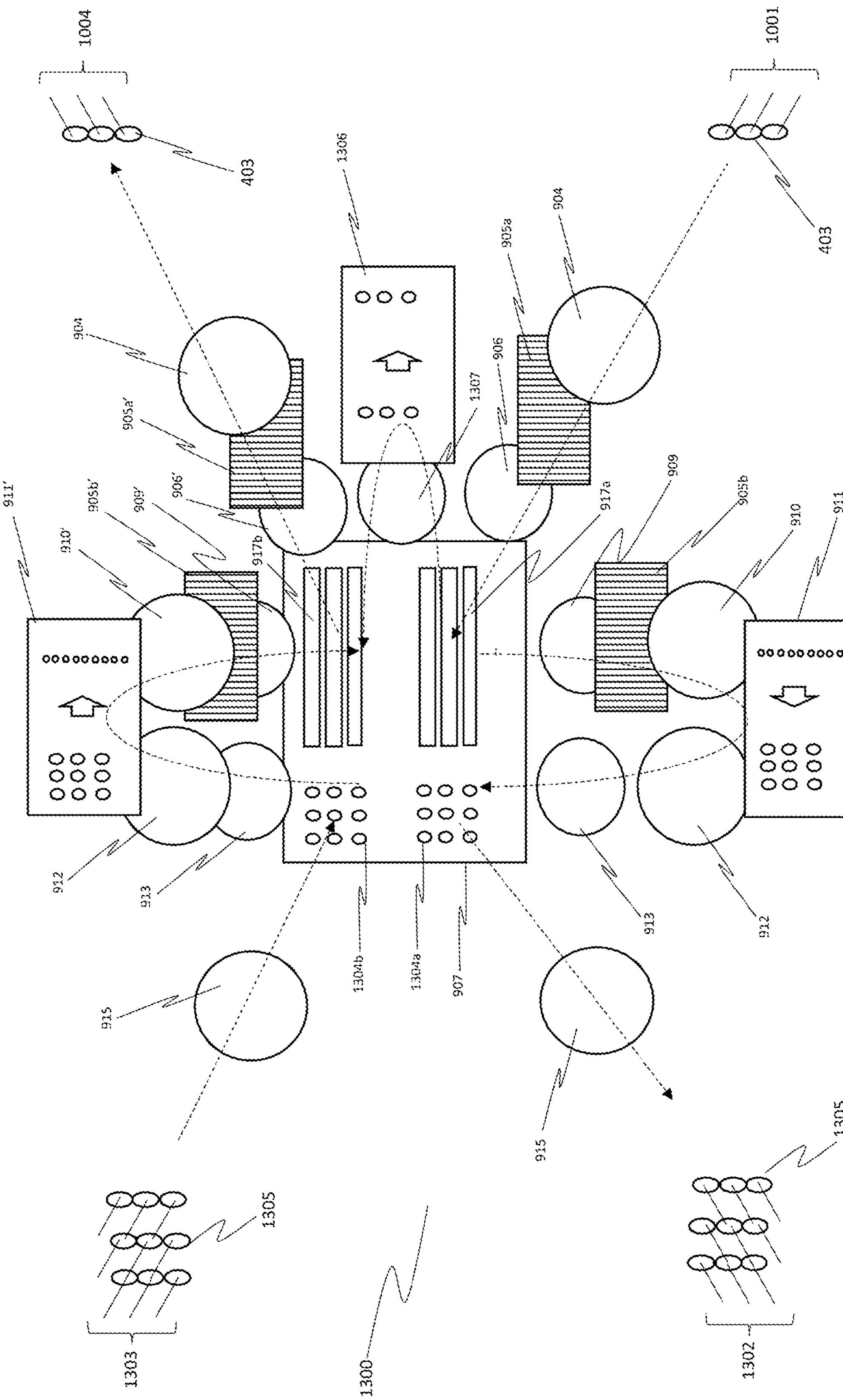

1004
1001
403
403
1306
904
905a
904
905a'
906
1307
911'
905b'
909'
906'
917b
917a
909
905b
910
911
910'
912
913
1304b
1304a
907
913
912
915
915
1305
1305
1303
1300
1302

Figure 13

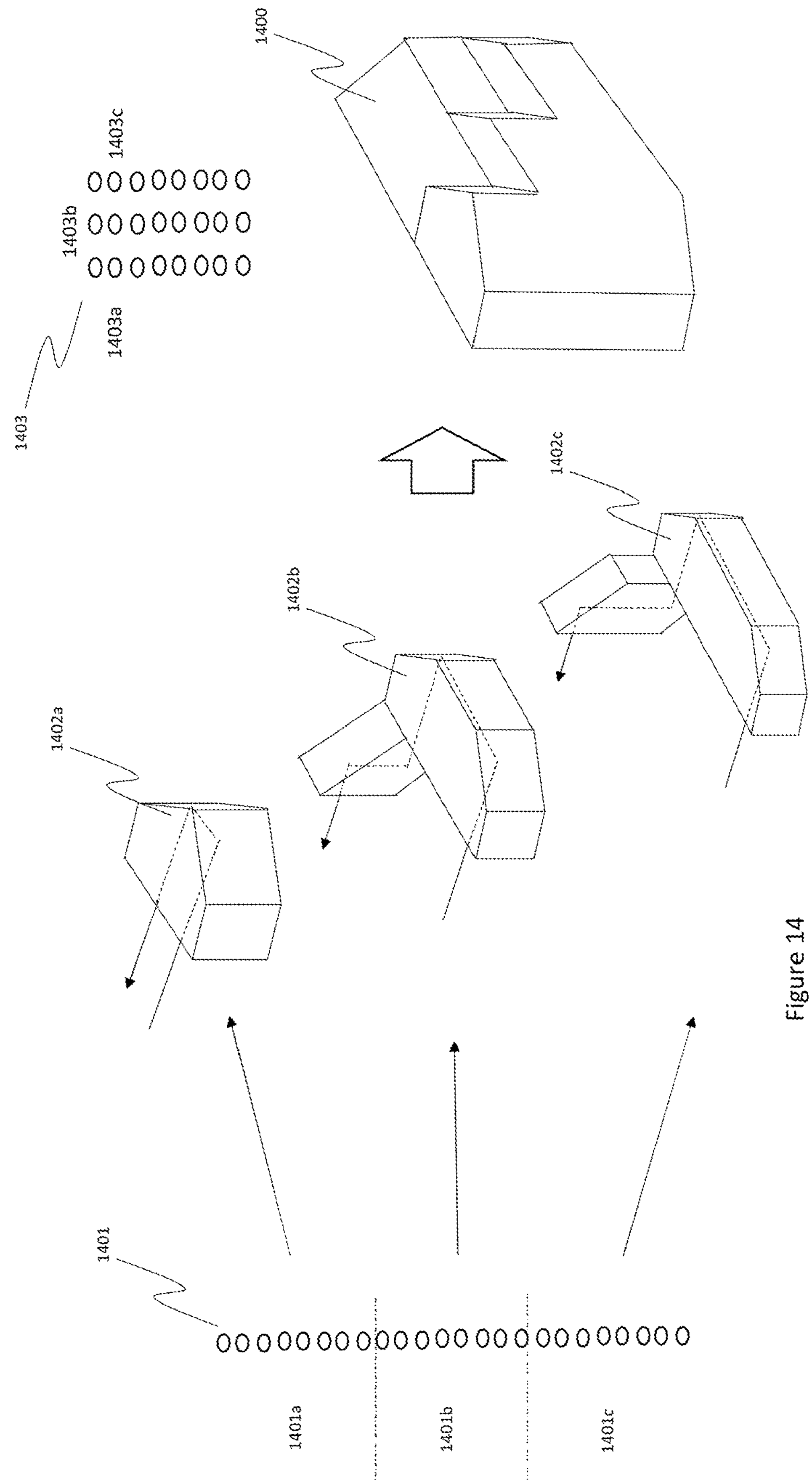

1400
1403c
1403b
1403a
1403
1402c
1402b
1402a
1401
1401a
1401b
1401c

Figure 14

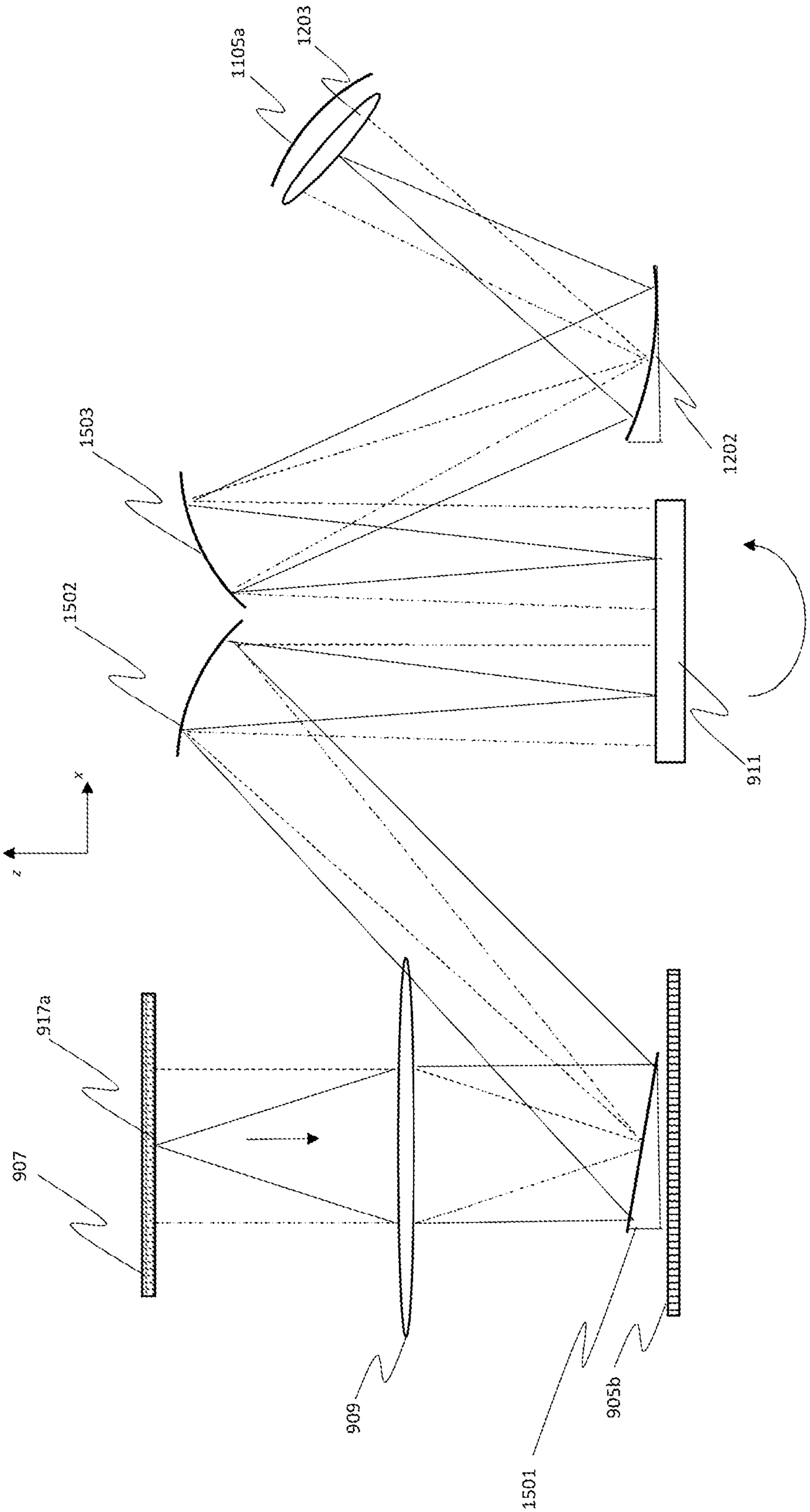


Figure 15

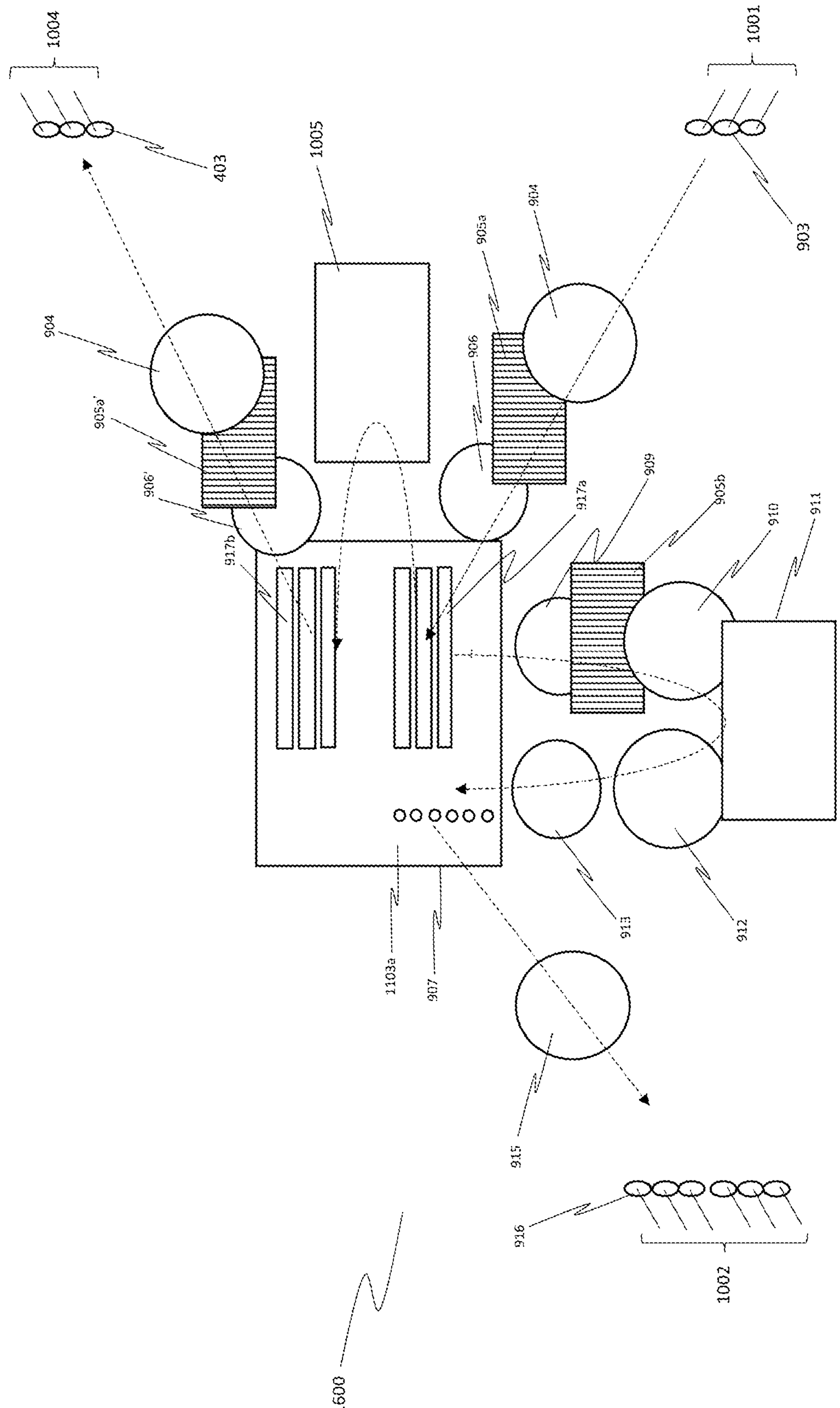

1004
403
1005
1001
903
904
905a
906
904
905a'
906'
917b
917a
909
905b
910
911
1103a
907
913
912
915
916
1002
1600

Figure 16

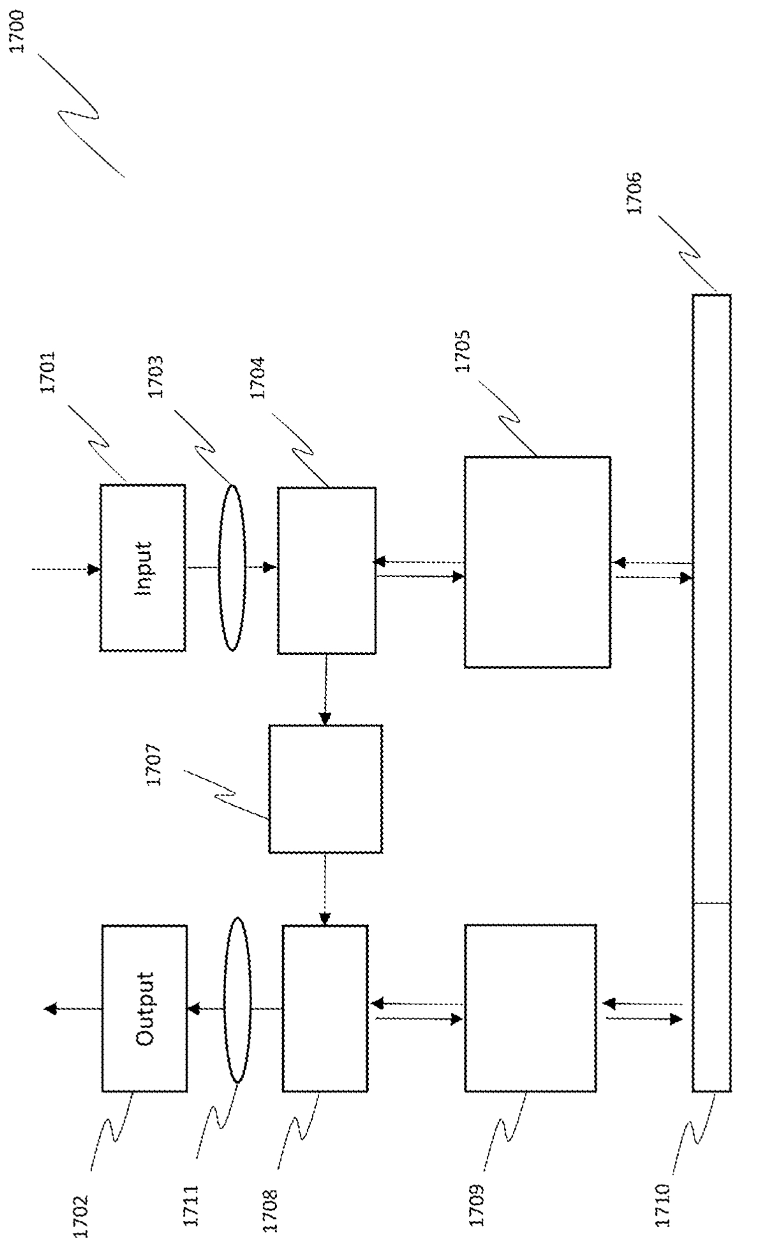

1700
1701
1703
1704
1705
1706
Input
1707
Output
1702
1711
1708
1709
1710

FIG. 17

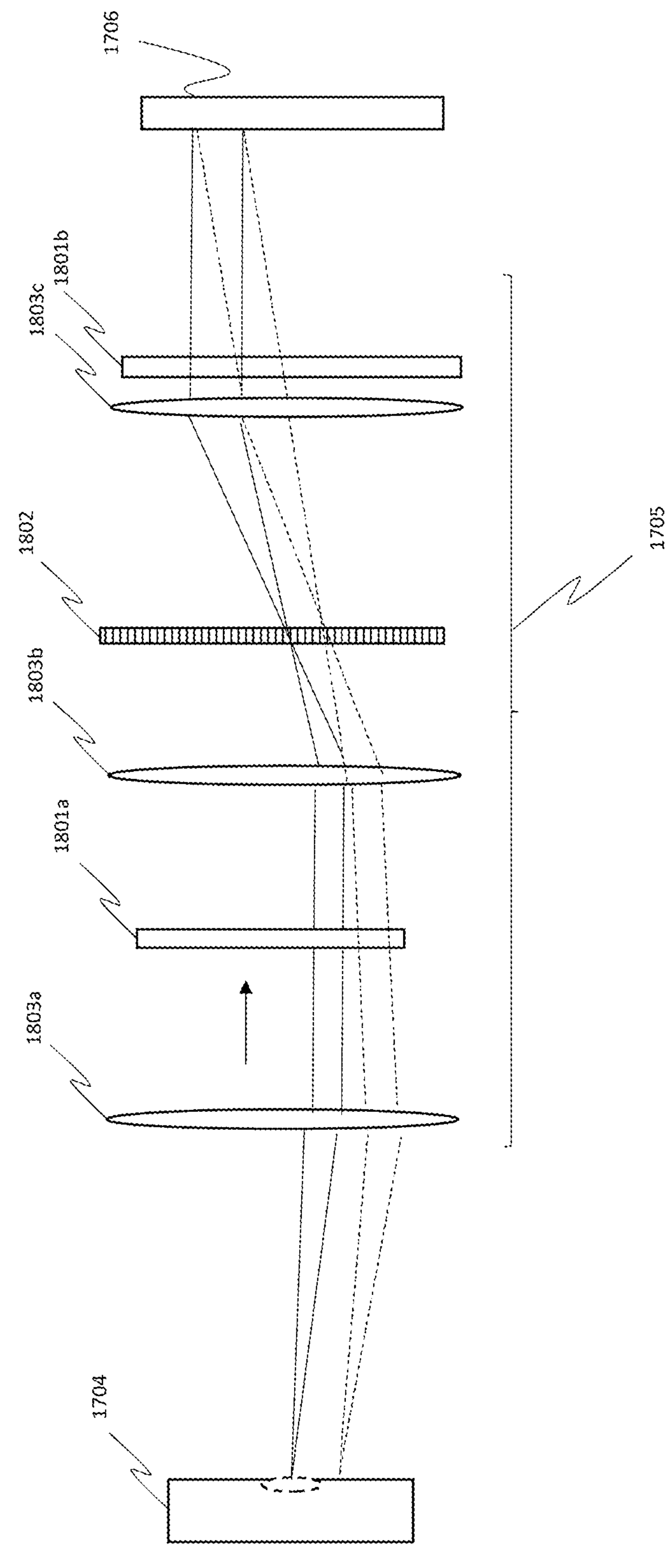


FIG. 18

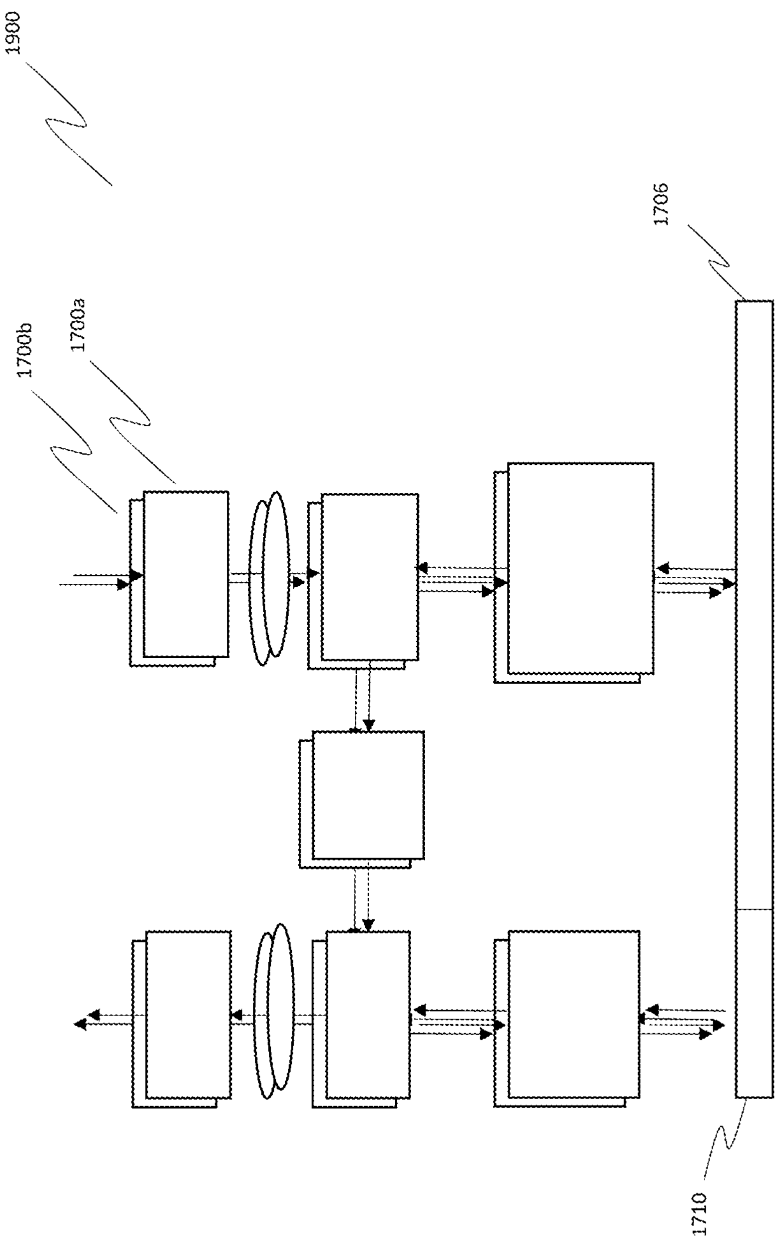


FIG. 19

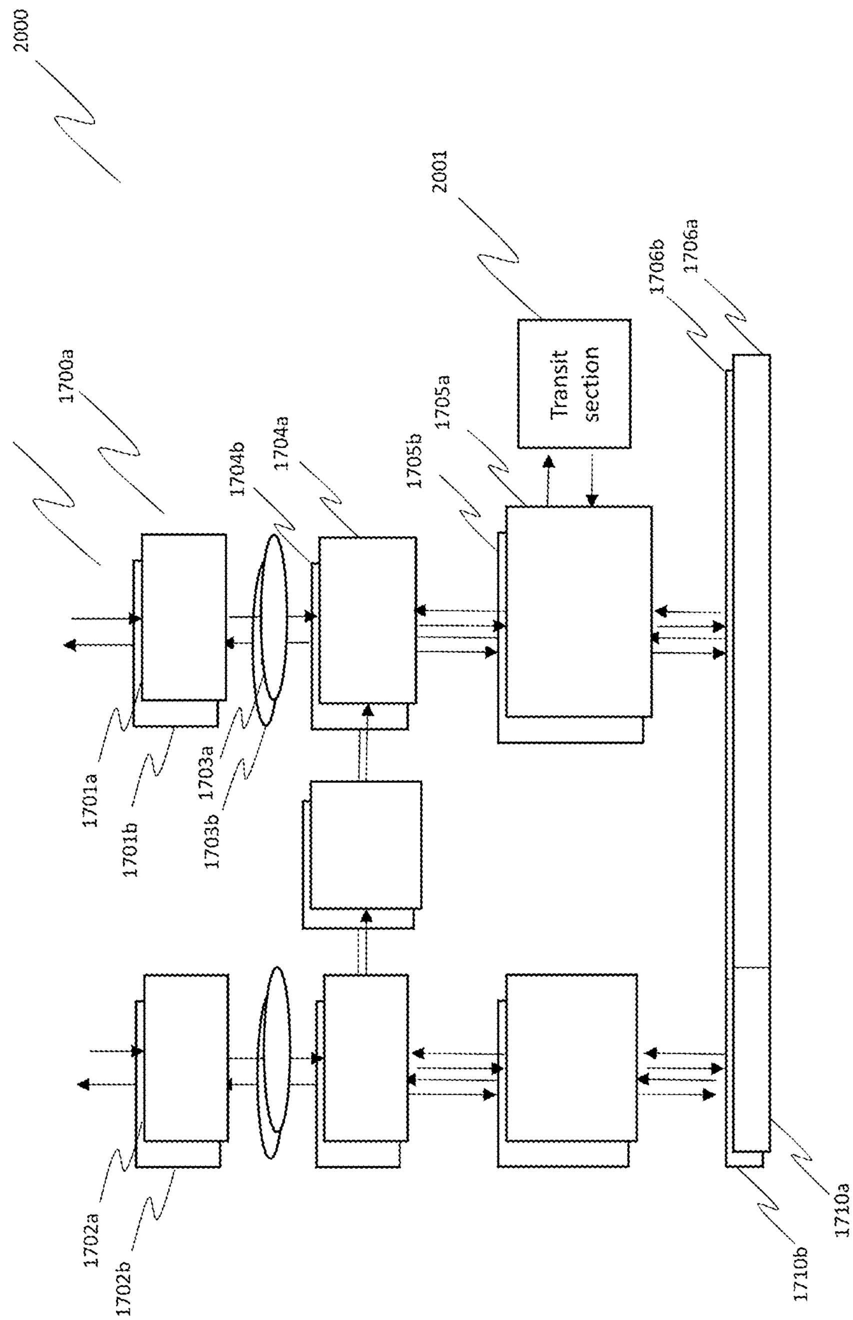

2000
2001
Transit section
1700a
1700b
1701a
1701b
1702a
1702b
1703a
1703b
1704a
1704b
1705a
1705b
1706a
1706b
1710a
1710b

FIG. 20

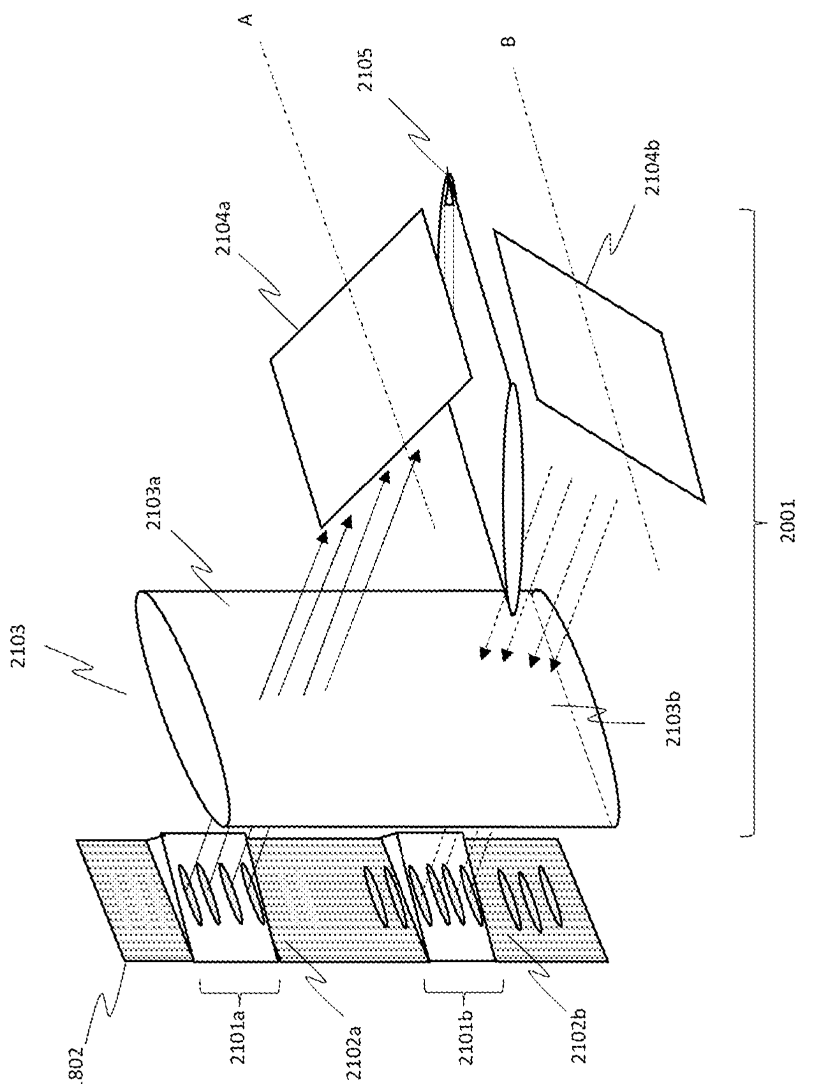

A
B
2105
2104a
2104b
2103a
2103
2103b
2001
1802
2101a
2102a
2101b
2102b

FIG. 21

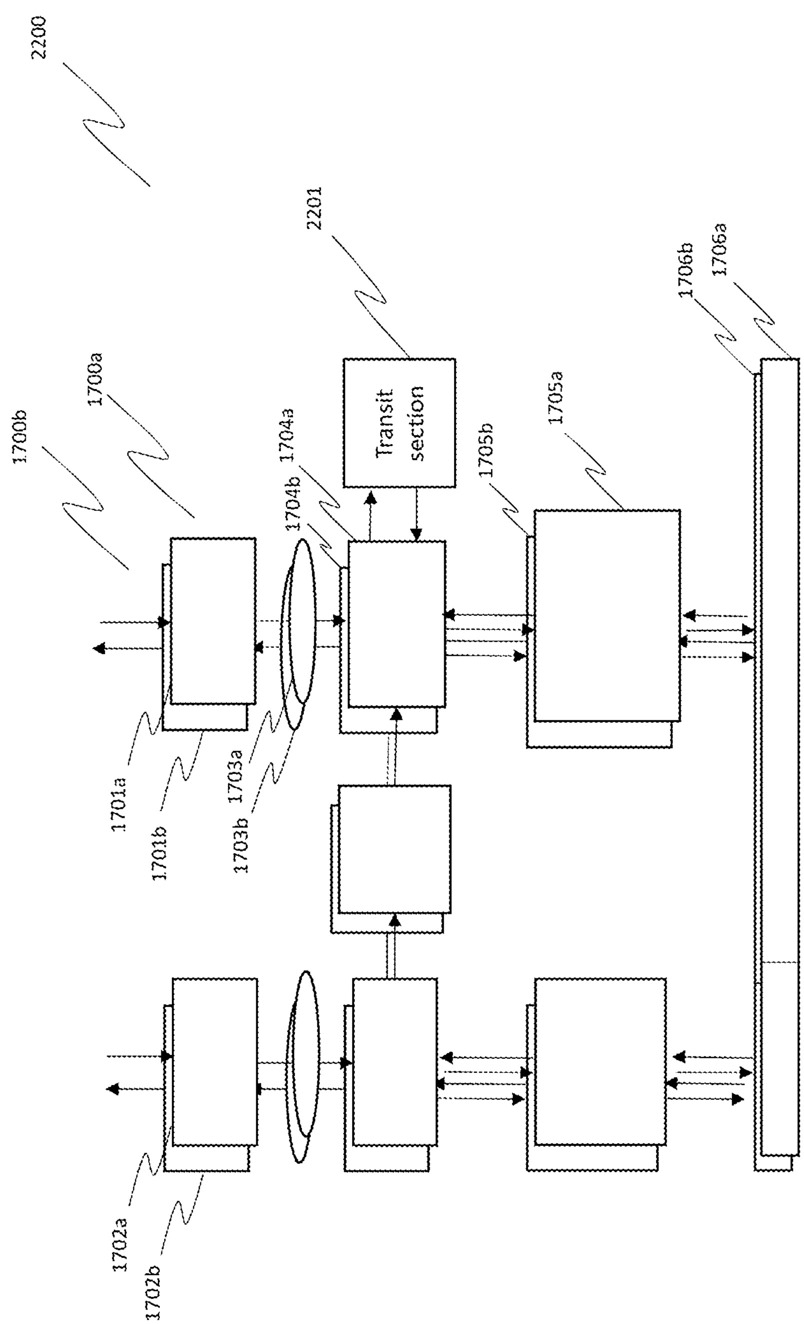
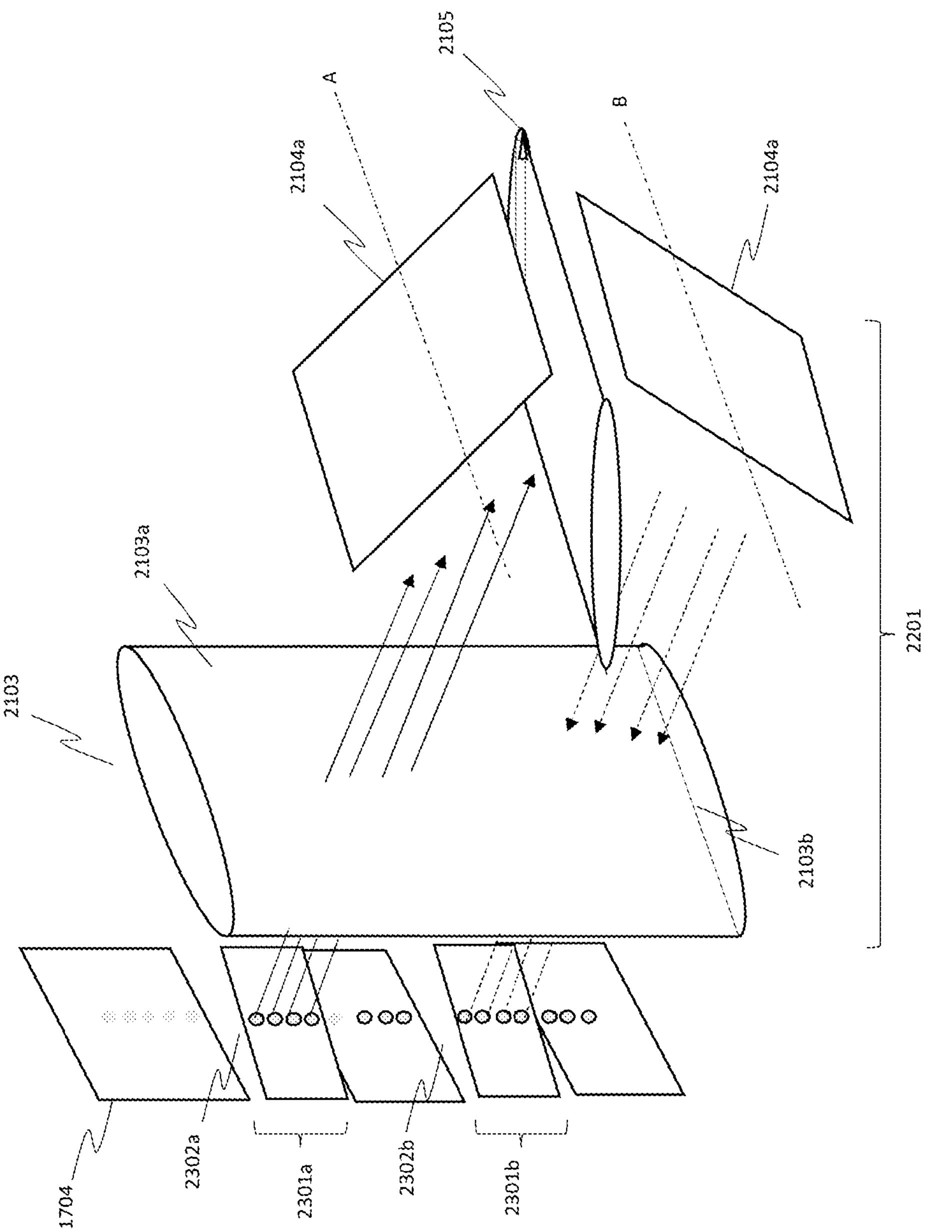

2200
2201
Transit section
1700b
1700a
1704a
1704b
1705b
1705a
1706b
1706a
1701a
1701b
1703a
1703b
1702a
1702b

FIG. 22



2105
A
B
2104a
2104a
2103a
2103
2201
2103b
1704
2302a
2301a
2302b
2301b

FIG. 23

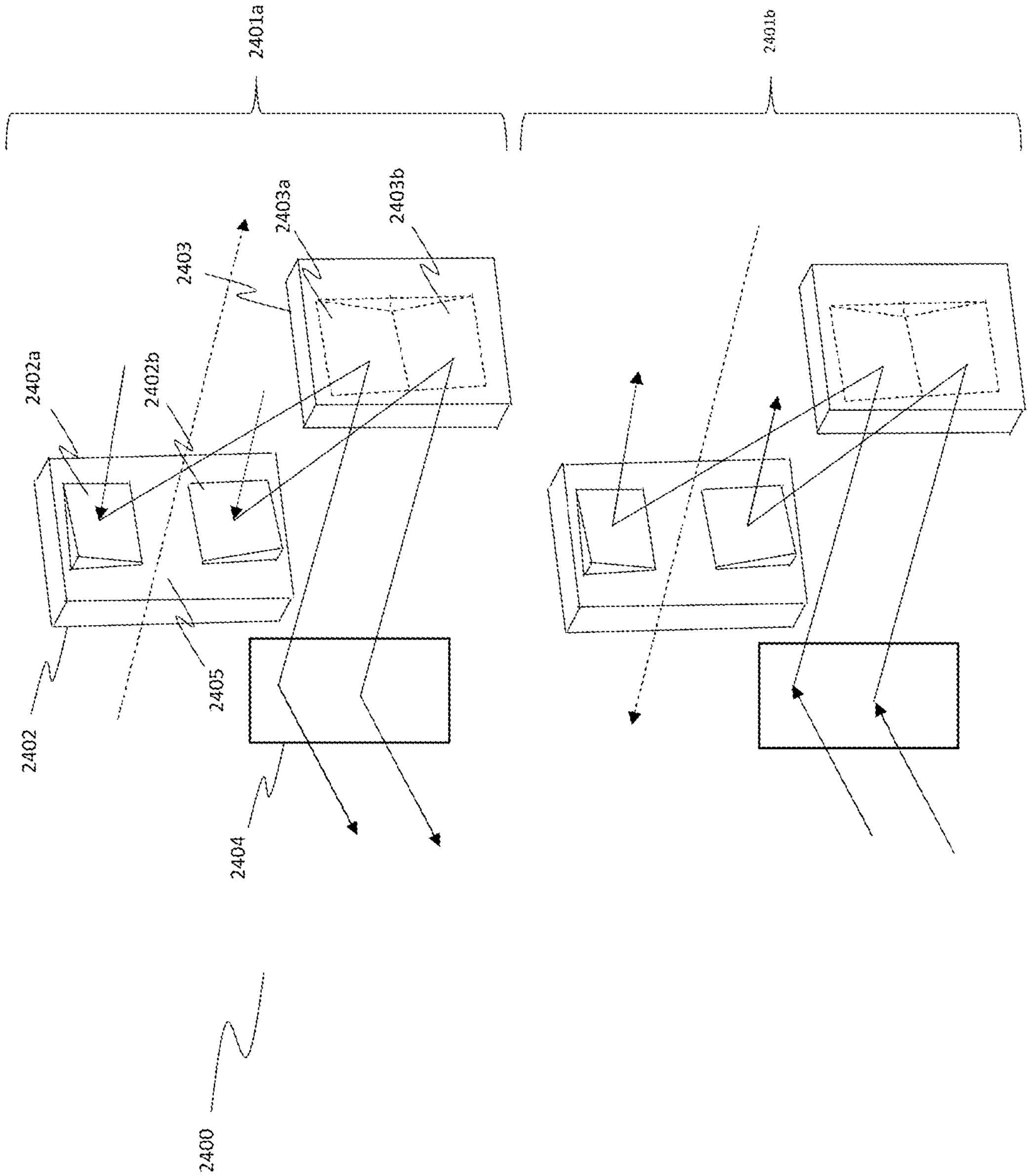

2401a
2401b
2402a
2402b
2403
2403a
2403b
2402
2405
2404
2400

FIG. 24

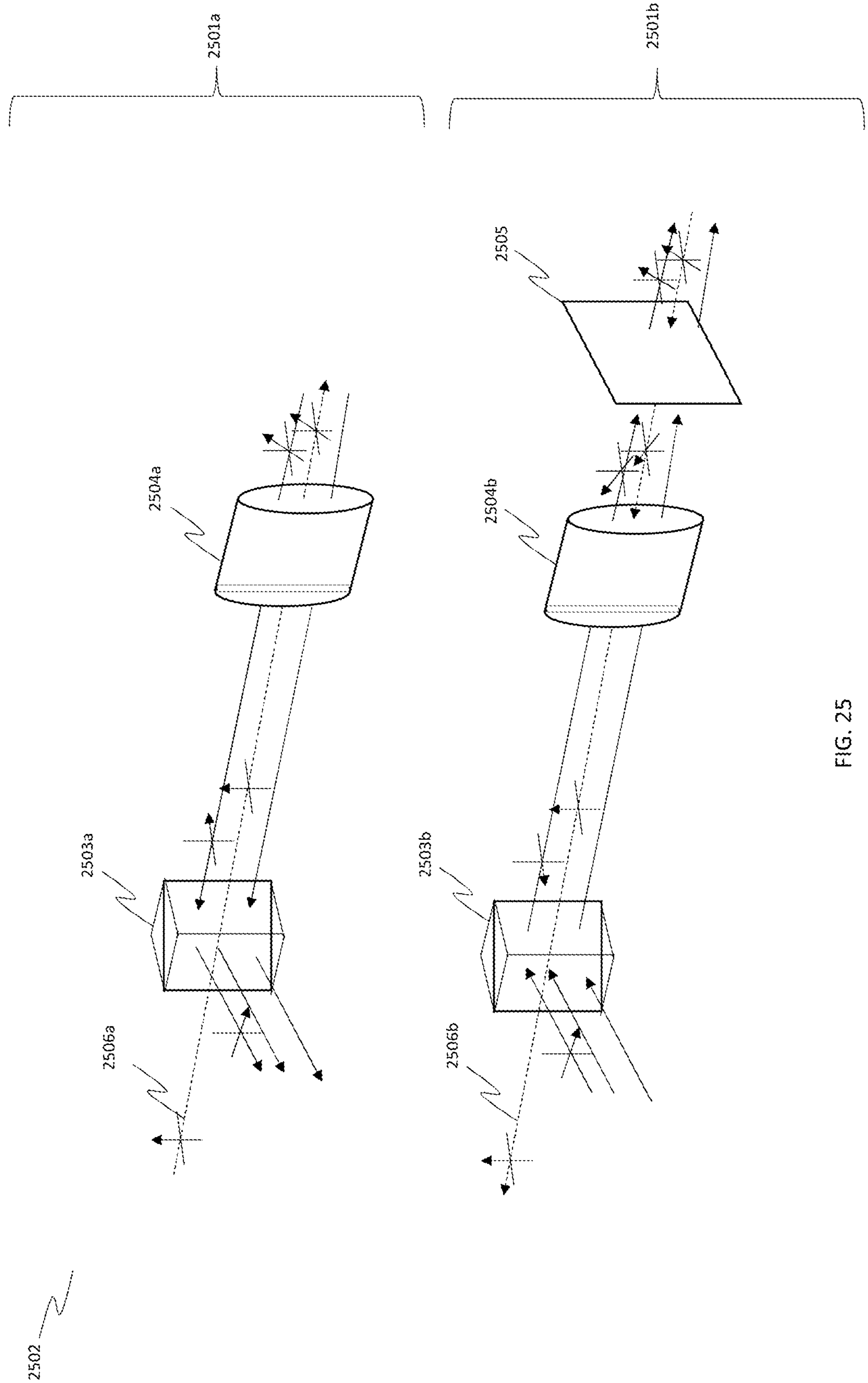


FIG. 25

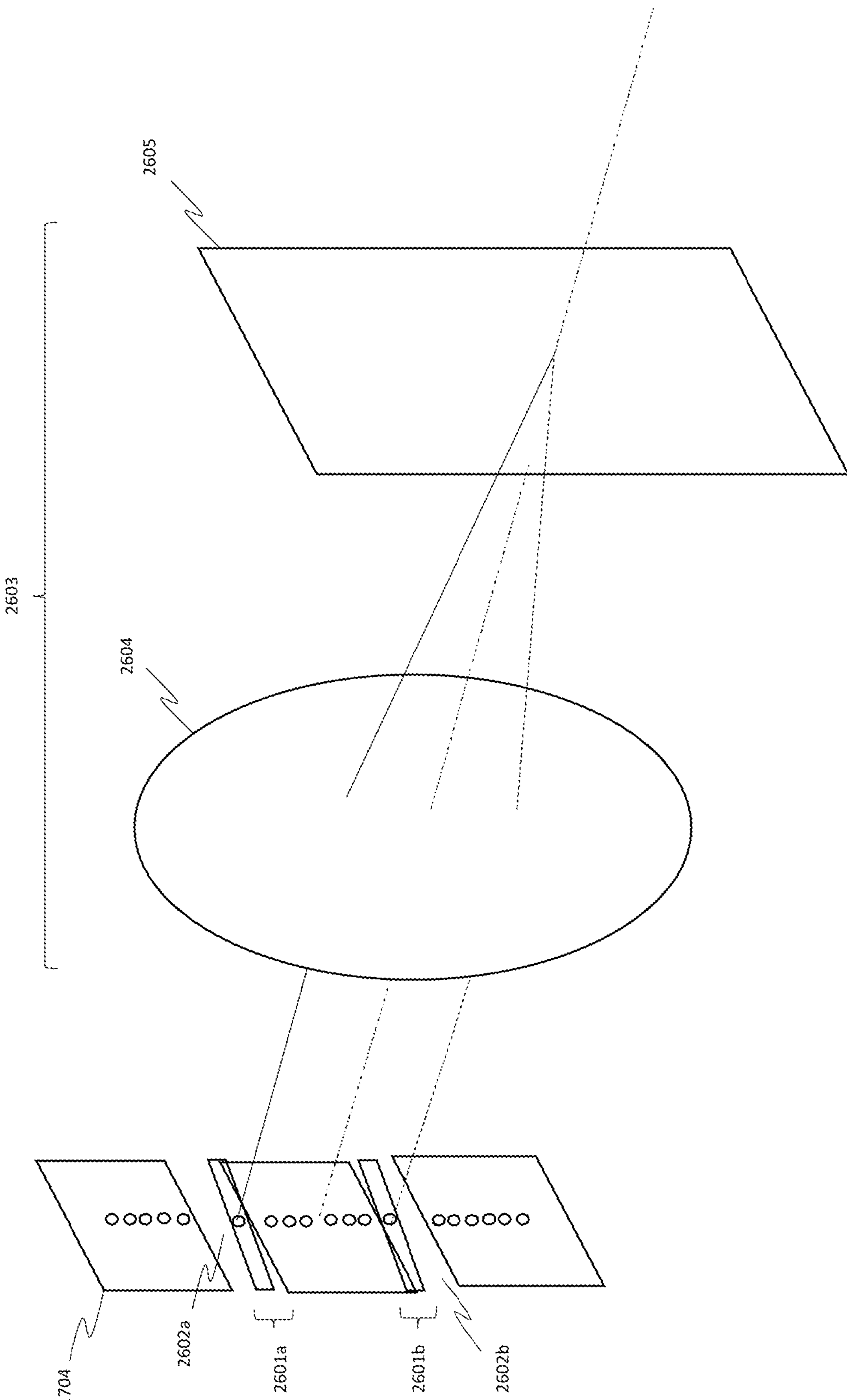


FIG. 26

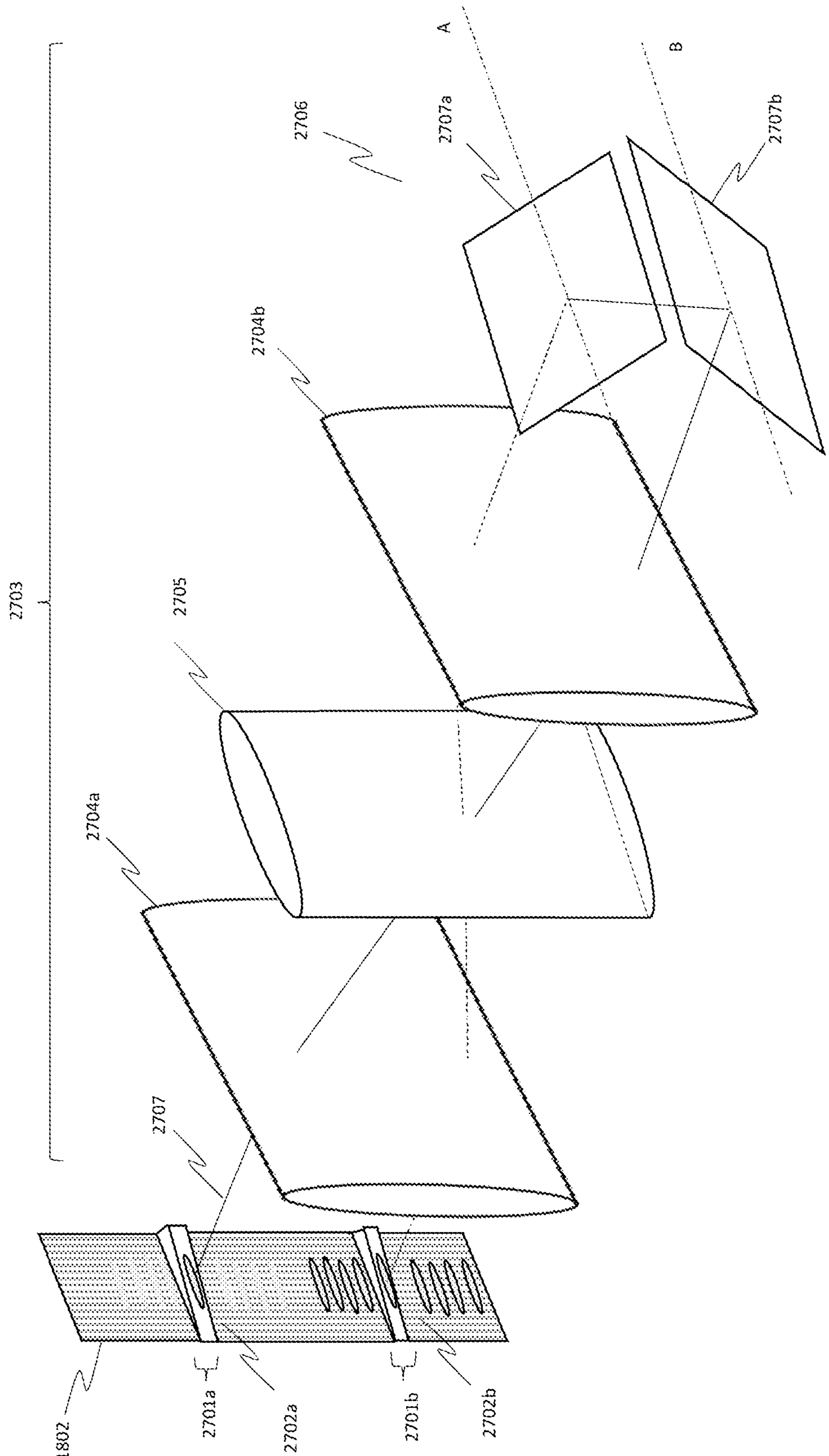

2703
2706
A
B
2707a
2707b
2704b
2705
2704a
2707
1802
2701a
2702a
2701b
2702b

FIG. 27

OPTICAL SWITCH

BACKGROUND

[0001] Optical switches are used in optical telecommunication systems to route optical signals through networks. As optical telecommunications systems have become more popular, the quantity of data carried through the networks has increased, putting greater capacity demands on the switches. It is known to use wavelength division multiplexed (WDM) signals to enable each optical fibre in the network to carry multiple data channels, those data channels separated by unique central frequencies and having non-overlapping bandwidths. Wavelength selective switches (WSSs) are used to route WDM signals through the network.

[0002] FIG. **1** illustrates schematically a known M×N WSS **100**. The M×N switch comprises N input ports **101** and M output ports **102**. Each port carries multiple data channels. A bank of 1×M WSSs **103** splits the multiplexed signal from each input port into its separate frequency channels. The demultiplexed data channels are then directed to the M output ports. A bank of N×1 WSSs **104** at the output combines the data channels into a set of multiplexed signals for output via the output ports **102**. In this way, the M×N switch is able to redirect any data channel from an input port to any data channel in an output port, subject to the condition that two channels with overlapping frequencies are not routed to the same output port.

[0003] FIG. **2** illustrates schematically a known switch referred to as an add-drop WSS **200**. Add-drop WSS **200** is a special type of WSS in which N input ports **201** are connected to K output ports **202**, where K>N. A bank of 1×K WSSs **203** splits the multiplexed signal from each input port into its separate frequency channels. The demultiplexed data channels are then directed to K space switches **204**. Each space switch **204** can accept data from any of the 1×K WSSs but can only output data from one of the input ports at a time. The output of each space switch **204** is then output from an output port **202**, otherwise known as a drop port. Space switches are simpler to implement than N×1 switches, and hence the add-drop WSS of FIG. **2** is preferable to the M×N WSS of FIG. **1**. This is particularly the case when K is much bigger than N, and the data density in the K output ports is much lower than in the N input ports. Both the WSSs of FIGS. **1** and **2** are reversible. For example, edge reconfigurable optical add-drop multiplexers (ROADM) are used for transferring optical data between core dense wavelength divisional multiplexing (DWDM) and more coarse wavelength division multiplexing (CWDM). An add-drop WSS of the type in FIG. **2** is used in the "dropping" direction shown in FIG. **2** to transfer from DWDM to CWDM, and in the reverse "adding" direction (from K drop input ports to N output ports) to transfer from CWDM to DWDM.

[0004] The switches shown in FIGS. **1** and **2** are typically implemented by electronic conversion, for example by absorbing light and emitting it again in the desired format. However, electronic conversion has high power requirements and introduces a significant lag in the transmission through the switch. Purely optical methods require much less power than electronic implementations and also enable faster transmission.

[0005] FIG. **3** illustrates a known optical implementation **300** of the M×N WSS of FIG. **1** or the add-drop WSS of FIG. **2**. N input ports **301** each carry multiple channels. Light through each input port **301** is directed to an imaging telescope **302** where it is focused to a point, and then imaged by a 4F optical system **303** and **305**. This produces parallel light beams which are then split into their constituent frequency channels by diffraction grating **304**. The dispersed channels are focused through lens **305** to a spatial light modulator (SLM) plane **306**, which is typically implemented by a liquid crystal on silicon (LCOS) device. The LCOS device applies a holographic beam deflection to the spectrum of channels incident on it to direct each channel from the N input spectra towards the M spectra on a second SLM plane **308**. The deflected channels pass through a Fourier lens (or lens group) **307** to form the conjugate plane before reaching the second SLM plane **308**. The second SLM plane **308** is typically an LCoS or optical microelectromechanical system (MEMS). The second SLM plane **308** applies an angle deflection to the light incident on it to direct it towards the output ports **309**. The M deflected spectra pass through a lens **310** to generate parallel light beams, which then travel through diffraction grating **311** to recombine the spectra for each of the M output ports **309**. The recombined spectra then pass through 4F optical system **312**. Fourier lens arrangement **314** is used to select the port which the multiplexed signals are then output through.

[0006] An undeviated beam **315** is aligned with the central output port **309***a*. An angular deflection to this beam channel **316** of a certain magnitude selects an alternative output port **309***b*. The deflection of the second SLM plane **308** corrects for the incidence angle from the different spectra on the first SLM plane **306** so that light from any of the input port spectra can be coupled to any of the output ports.

[0007] For the M×N WSS implementation, the second SLM plane **308** of FIG. **3** is a pixelated SLM such as an LCoS. For the add-drop WSS implementation, the second SLM plane **308** of FIG. **3** can be a MEMS array that is pixelated in one direction only (vertical) in order to act as the space switches. In this case, the second SLM plane **308** may be before or after the second diffraction grating **311**.

[0008] These known optical WSSs utilise two separate SLM devices, one for each of the SLM planes **306**, **308**. SLM devices are expensive and limit the compactness of the WSS. Even with the simpler add-drop WSS, the switching is between a vertical column of input spectra on the first SLM plane **306** to a vertical column of space switch areas on the second SLM plane **308**. The large deflection angle on the two SLM planes required to switch light from the top spectra on the first SLM plane **306** to the bottom space switch on the second SLM plane **308** limits the number of accessible output ports M that can be obtained with this arrangement, and hence limits the capacity of a switch having this type of architecture.

[0009] Two adWSS structures may be incorporated into a two layer edge reconfigurable optical add-drop multiplexer (ROADM) wavelength selective switch, such as that described in US 2022/0052778. Four SLM planes are required for each ROADM switch. US 2022/0052778 utilises opposing SLM planes as part of its N×N switch, which necessitates utilisation of separate SLM devices.

SUMMARY OF THE INVENTION

[0010] According to an aspect of the invention, there is provided an optical switch comprising: a set of first input ports, each first input port configured to transport an optical signal having at least one component frequency channel; a set of first output ports, each first output port configured to transport an optical signal having at least one component frequency channel; a first programmable deflection plane configured to deflect beams incident on it to form a first deflected array of beams; a second programmable deflection plane configured to deflect beams incident on it to form a second deflected array of beams; a first demultiplexer configured to separate light from the set of first input ports into its component frequency channels to form the beams incident on the first programmable deflection plane; a first multiplexer configured to combine the second deflected array of beams from the second programmable deflection plane into combined signals incident on the set of first output ports; and a beam steering optical element group configured to transfer the first deflected array of beams from the first programmable deflection plane to the beams incident on the second programmable deflection plane; wherein the incidence normal of the first programmable deflection plane is within the same hemispherical angular area as the incidence normal of the second programmable deflection plane; and wherein the dispersion direction in the dispersion plane of spectra of the beams formed on the second programmable deflection plane with respect to the diffraction order of the first multiplexer is such that the combined signals from the first multiplexer match the set of first output ports.

[0011] The dispersion direction in the dispersion plane of the first multiplexer may be the same sign as the dispersion direction in the dispersion plane of the first demultiplexer.

[0012] The first multiplexer may be the same as the first demultiplexer.

[0013] The beam steering optical element group may be configured to form a non-inverted image of the first programmable deflection plane on the second programmable deflection plane in the dispersion plane.

[0014] The beam steering optical element group may be configured to form a Fourier conjugate image of the first programmable deflection plane on the second programmable deflection plane orthogonal to the dispersion plane.

[0015] The beam steering optical element group may comprise: a first optical element and a second optical element, the first and second optical elements configured to divert the first deflected array of beams from the first programmable deflection plane towards a beam steering optical device; the beam steering optical device configured to redirect the diverted first deflected array of beams to form a redirected first deflected array of beams; and a third optical element and a fourth optical element, the third and fourth optical elements configured to divert the redirected first deflected array of beams from the beam steering optical device towards the second programmable deflection plane.

[0016] Each of the first optical element, second optical element, third optical element, fourth optical element and beam steering optical device may have optical power.

[0017] The dispersion direction in the dispersion plane of the first multiplexer may be the opposite sign as the dispersion direction in the dispersion plane of the first demultiplexer.

[0018] The beam steering optical element group may be configured to form an inverted image of the first programmable deflection plane on the second programmable deflection plane in the dispersion plane.

[0019] The beam steering optical element group may be configured to form a Fourier conjugate image of the first programmable deflection plane on the second programmable deflection plane orthogonal to the dispersion plane.

[0020] The beam steering optical element group may comprise: a first optical element configured to divert the first deflected array of beams from the first programmable deflection plane towards a beam steering optical device; the beam steering optical device configured to redirect the diverted first deflected array of beams to form a redirected first deflected array of beams; and a second optical element configured to divert the redirected first deflected array of beams from the beam steering optical device towards the second programmable deflection plane.

[0021] Each of the first optical element, second optical element and beam steering optical device may have optical power.

[0022] The beam steering optical element group may comprise a lens assembly.

[0023] The lens assembly may comprise a spherical lens and a cylindrical lens with power only in the dispersion axis of the cylindrical lens.

[0024] The optical switch may further comprise: a first mirror segment configured to divert a portion of the first deflected array of beams from the first programmable deflection plane to a curved mirror; and a second mirror segment configured to divert the portion of the first deflected array of beams from the curved mirror to the second programmable deflection plane.

[0025] The optical switch may further comprise: a first mirror segment, a second mirror segment and a further beam steering optical device; the first mirror segment configured to divert a portion of the first deflected array of beams from the first programmable deflection plane to the further beam steering optical device; the further beam steering optical device configured to remap the portion of the first deflected array of beams to form a remapped portion of beams directed to the second mirror segment; and the second mirror segment configured to divert the remapped portion of beams from the further beam steering optical device to the curved mirror, and from the curved mirror to the second programmable deflection plane.

[0026] The optical switch may comprise a single spatial light modulator (SLM) panel comprising the first and second programmable deflection planes.

[0027] According to a further aspect of the invention, there is provided a reconfigurable optical add-drop multiplexer (ROADM) optical switch comprising the optical switch described above, and further comprising: a set of second input ports, each second input port configured to transport an optical signal having at least one component frequency channel; a set of second output ports, each second output port configured to transport an optical signal having at least one component frequency channel; a third programmable deflection plane configured to deflect beams from the set of second input ports incident on it to form a third deflected array of beams; a fourth programmable deflection plane configured to deflect beams incident on it to form a fourth deflected array of beams towards the set of second output ports; a second demultiplexer configured to separate light from the third programmable deflection plane into its component frequency channels towards the second programmable deflection plane; a second multiplexer configured to combine the first deflected array of beams from the first programmable deflection plane into combined signals towards the fourth programmable deflection plane; wherein the incidence normal of each of the first, second, third and fourth programmable deflection planes is within the same hemispherical angular area; and wherein the first demultiplexer and second multiplexer have the same dispersion direction; and wherein the second demultiplexer and first multiplexer have the same dispersion direction.

[0028] The first demultiplexer may be the same as the second multiplexer.

[0029] The second demultiplexer may be the same as the first multiplexer.

[0030] A single spatial light modulator (SLM) panel may comprise the first, second, third and fourth programmable deflection planes.

[0031] The switch may further comprise a mirror optic, the mirror optic comprising a pair of mirrors separated by a gap.

[0032] The switch may comprise a further mirror optic, the further mirror optic comprising a pair of mirrors separated by a gap.

[0033] The first lens assembly may comprise the first demultiplexer.

[0034] The first lens assembly may comprise a further two lenses having optical power along the respective steering axes of the two lenses.

[0035] The beam steering optical element group may comprise a second lens assembly.

[0036] The second lens assembly may comprise a spherical lens and a cylindrical lens with power only in the dispersion axis of the cylindrical lens.

[0037] The second lens assembly may comprise the first multiplexer.

[0038] The second lens assembly may comprise a further two lenses having optical power along the respective steering axes of the two lenses.

[0039] The beam steering optical device may comprise a lens and mirror assembly.

[0040] The lens and mirror assembly may comprise a cylindrical lens with optical power in the dispersion axis of the cylindrical lens, a pair of reflecting mirrors and a lens with optical power in the steering axis of the lens.

[0041] The optical path between the first programmable deflection plane and the second programmable deflection plane may comprise five elements with optical power.

[0042] The optical path between the first programmable deflection plane and the second programmable deflection plane may comprise the beam steering optical device, the spherical lens of the first lens assembly and the spherical lens of the second lens assembly.

[0043] The beam steering optical device may be configured to transfer demultiplexed beams.

[0044] The first demultiplexer may comprise at least one transit input port. The first multiplexer may comprise at least one transit output port. The first demultiplexer may be configured to direct the first deflected array of beams to the beam steering optical device. The first multiplexer may be configured to direct the first deflected array of beams from the beam steering optical device towards the second programmable deflection plane.

[0045] The optical path between the first programmable deflection plane and the second programmable deflection plane may comprise nine elements with optical power.

[0046] The optical path between the first programmable deflection plane and the second programmable deflection plane may comprise the beam steering optical device, the first lens assembly and the second lens assembly.

[0047] The beam steering optical device may be configured to transfer multiplexed beams.

[0048] The mirror optic may comprise at least one transit input port. The further mirror optic may comprise at least one transit output port. The mirror optic may be configured to direct the first deflected array of beams to the beam steering optical device and the further mirror optic may be configured to direct the first deflected array of beams from the beam steering optical device towards the second programmable deflection plane.

[0049] The first optical element configured to divert the first deflected array of beams from the first programmable deflection plane towards the beam steering optical device may be further configured to alter the configuration of the first deflected array of beams so as to generate or remove a gap between the first deflected array of beams.

[0050] The optical path between the first programmable deflection plane and the second programmable deflection plane may comprise a polarising beam splitter and a Faraday rotator.

BRIEF DESCRIPTION OF THE FIGS.

[0051] The present invention will now be described by way of example with reference to the accompanying drawings. In the drawings:

[0052] FIG. **1** illustrates a known M×N WSS;

[0053] FIG. **2** illustrates a known add-drop WSS;

[0054] FIG. **3** illustrates a known implementation of an M×N or add-drop WSS;

[0055] FIG. **4** illustrates an M×N WSS;

[0056] FIG. **5** illustrates two programmable deflection planes at an angle to each other;

[0057] FIG. **6** illustrates programmable deflection planes being virtually relocated by local optics;

[0058] FIG. **7** illustrates an M×N WSS;

[0059] FIG. **8** illustrates the optical path of an exemplary beam steering optical element group comprising five optical elements;

[0060] FIG. **9** illustrates an add-drop WSS;

[0061] FIG. **10** illustrates an edge ROADM;

[0062] FIG. **11** illustrates the optical path of a transit section of the edge ROADM of FIG. **10**;

[0063] FIGS. **12***a* and **12***b* illustrate the optical path of a transit section of the edge ROADM of FIG. **10**;

[0064] FIG. **13** illustrates an edge ROADM;

[0065] FIG. **14** illustrates a prism array implementation of a beam steering optical device;

[0066] FIG. **15** illustrates the optical path of a transit section of the edge ROADM where the spectra are remapped;

[0067] FIG. **16** illustrates an optical switch utilising three programmable deflection planes.

[0068] FIG. **17** illustrates a further M×N WSS;

[0069] FIG. **18** illustrates the optical path through an optical system of the M×N WSS;

[0070] FIG. **19** illustrates an edge ROADM;

[0071] FIG. **20** illustrates an edge ROADM;

[0072] FIG. **21** illustrates a demultiplexer and a transit section;

[0073] FIG. **22** illustrates an edge ROADM;

[0074] FIG. **23** illustrates a mirror optic and a transit section;

[0075] FIG. **24** illustrates another mirror optic and a transit section;

[0076] FIG. **25** illustrates an alternative optical arrangement configured to direct light to and receive light from a transit section;

[0077] FIG. **26** illustrates a mirror optic and a transit section; and

[0078] FIG. **27** illustrates a demultiplexer and a transit section

DETAILED DESCRIPTION

[0079] The following describes several exemplary optical switches which utilise two or more programmable deflection planes and a beam steering optical element group to optically route light from a set of input ports to a set of output ports. The described optical switches may be

[0080] WSSs. These WSSs may be implemented as add-drop WSSs (adWSS), for example for transferring light from core DWDM networks to lower capacity CWDM networks. Alternatively, the described WSSs may be implemented as M×N WSSs. All the examples described herein use optical components to route the light through the switches. There is no absorption and re-emission of light.

[0081] FIG. **4** illustrates an M×N WSS **400** in which a single panel **407** comprises two programmable deflection planes **408**, **414**. This single panel may, for example, be an SLM device. Both the programmable deflection planes **408**, **414** are spectral planes configured to deflect each beam of a dispersed optical signal individually.

[0082] In the M×N WSS of FIG. **4**, N input ports **401** are routed to M output ports **402**. Each port may comprise one or more optical fibres. Each port transports an optical signal having at least one component data channel. Typically, each port has a plurality of data channels. Each channel transports data. Each channel has a frequency band which has a different centre frequency to the other channels of that port. The channels of a port may have the same or different bandwidths. The bandwidths of the channels of a port are non-overlapping. As will be described below, each channel from each input port **401** may be routed to any individual output port **402**.

[0083] The optical signals from the input ports **401** pass through coupling lenses **403** which form an optical beam of the N ports before a 4F imaging system. There is a coupling lens for each input port. Optical signals from the input ports **401** pass through a 4F imaging system provided by lenses **404** and **406**. The optical signals are dispersed by diffraction grating **405*a*** at the Fourier plane of lens **404**. The diffraction grating **405*a*** spatially separates the component frequency channels of the optical signal beams incident on it to form a set of dispersed optical signal beams. In other words, each optical signal from an input port is spread into its frequency spectrum by the diffraction grating **405*a***. Thus, the diffraction grating **405*a*** acts to demultiplex each optical signal into its component data channels.

[0084] A first programmable deflection plane **408** is positioned in the optical path of the set of dispersed optical signals such that the set of dispersed optical signal beams form spectra **417** on the first programmable deflection plane **408**. In the example of FIG. **4**, three input ports **401** are illustrated, each of which forms a spectrum **417** on the first programmable deflection plane **408**. Each spectrum forms an array of beams of the component data channels of the input port which are imaged on the first programmable deflection plane **408**, each beam being a data channel. The first programmable deflection plane **408** applies a programmable deflection to each beam of the array of beams individually to form a deflected array of beams. The degree of deflection applied to each beam is so as to change the angle of that beam to direct its data channel on a path towards a selected one of the output ports **402**. The deflection applied by the first programmable deflection plane **408** is reconfigurable. Thus, a controller providing a control signal to the first programmable deflection plane **408** may change the deflection applied by the first programmable deflection plane **408** to each individual beam of the spectra from each input port, so as to output the desired channels to the desired ports.

[0085] The deflected array of beams from the first programmable deflection plane **408** is incident on a beam steering optical element group **409**. The beam steering optical element group **409** transfers the deflected array of beams from the first programmable deflection plane **408** onto the second programmable deflection plane **414**. The beam steering optical element group **409** forms a non-inverted image of the first programmable deflection plane on the second programmable deflection plane in the dispersion plane. In other words, if the spectra on the first programmable deflection plane increase in wavelength from left to right across the first programmable deflection plane, then the spectra on the second programmable deflection plane also increase in wavelength from left to right across the second programmable deflection plane. On the other hand, if the spectra on the first programmable deflection plane decrease in wavelength from left to right across the first programmable deflection plane, then the spectra on the second programmable deflection plane also decrease in wavelength from left to right across the second programmable deflection plane. The beam steering optical element group **409** forms a Fourier conjugate image of the first programmable deflection plane on the second programmable deflection plane in a direction orthogonal to the dispersion plane.

[0086] In FIG. **4**, the beam steering optical element group **409** comprises five optical elements: lenses **410**, **411**, **413** and **415** and beam steering optical device **412**. Lenses **410** and **411** divert the deflected array of beams from the first programmable deflection plane **408** towards the beam steering optical device **412**. The lenses **410** and **411** image the deflected spectra **417** onto the beam steering optical device **412**. Lenses **410** and **411** are equal in focal length and are positioned one focal length apart from each other. Lens **410** is one focal length of lens **410** from the first programmable deflection plane. Lens **411** is one focal length of lens **411** from the beam steering optical device **412**.

[0087] The beam steering optical device **412** redirects the diverted deflected array of beams towards the second programmable deflection plane **414**. The beam steering optical device **412** is reflecting and in an imaging configuration. In other words, the beam steering optical device **412** has a focal length equal to half the focal length of each of lenses **410**, **411**, **413** and **415**. The beam steering optical device **412** may be, for example, a curved mirror.

[0088] Lenses **413** and **415** divert the redirected array of beams from the beam steering optical device **412** towards the second programmable deflection plane. Lenses **413** and **415** are equal in focal length and are positioned one focal length apart from each other. Lens **413** is one focal length of lens **413** from the beam steering optical device **412**. Lens **415** is one focal length of lens **415** from the second programmable deflection plane.

[0089] In the plane orthogonal to the dispersion axis, the lenses **410**, **411**, **413** and **415** along with the beam steering optical device **412** are in a Fourier configuration between the first and second programmable deflection planes. For example, the lenses **410**, **411**, **413** and **415** may have no optical power in the axis orthogonal to the dispersion axis, whilst the beam steering optical device **412** has optical power in the axis orthogonal to the dispersion axis that is equal to its distance from the first and second programmable deflection planes. In this example, the beam steering optical device **412** may be implemented as a compound lens formed of two cylindrical lenses.

[0090] Suitably, the focal length of lenses **410** and **411** is the same as the focal length of lenses **413** and **415**.

[0091] Each of the optical elements in the beam steering optical element group has optical power.

[0092] The second programmable deflection plane **414** deflects the light incident on it such that it is deviated in a common direction. Thus, the second programmable deflection plane **414** applies a programmable deflection to each spatially separated component frequency channel beam of the array of beams incident on it to form a spatially separated output array of beams. The deflection applied by the second programmable deflection plane **414** is reconfigurable. Thus, a controller providing a control signal to the second programmable deflection plane **414** may change the deflection applied by the second programmable deflection plane **414** to each beam from the beam steering optical element group **409** so as to output the desired channels to the desired ports.

[0093] Lenses **417** and **418** are positioned in a 4F arrangement between the second programmable deflection plane **414** and the output ports **402**. A diffraction grating **405***b* is positioned in the Fourier plane between lenses **417** and **418**. The spatially separated output array of beams deflected by the second programmable deflection plane **414** passes through the lenses **417** and **418** and diffraction grating **405***b*. Diffraction grating **405***b* recombines the spatially separated output array of beams for output from the switch. Thus, the diffraction grating **405***b* acts to multiplex the spectra to form a multiplexed array of beams for output from the switch **400**. The multiplexed signals are routed to the lens array **419** by the lenses **417** and **418**. The lens array **419** couples the multiplexed signals to the output ports **402**.

[0094] Suitably, the diffraction grating **405***a* and the diffraction grating **405***b* have the same order. Suitably, the dispersion direction in the dispersion plane of the diffraction grating **405***b* is the same sign as the dispersion direction in the dispersion plane of the diffraction grating **405***a*.

[0095] The diffraction grating **405***b* may be the same as the diffraction grating **405***a*. The diffraction grating **405***b* may be different to the diffraction grating **405***a*.

[0096] In the example of FIG. **4**, the incidence normal of the first programmable deflection plane **408** is in the same direction as the incidence normal of the second programmable deflection plane **414**. This is because the first and second programmable deflection planes are coplanar. The first and second programmable deflection planes are located on the same single planar SLM device. In FIG. **4** an odd number of separated optical elements in the beam steering optical element group **409** (in this case five) is used in order for a Fourier conjugate image of the first programmable deflection plane **408** to be formed on the second programmable deflection plane **414** in a direction orthogonal to the dispersion plane.

[0097] Whilst also odd numbers of separated optical elements, one or three separated optical elements are insufficient to control the frequency channels from the second programmable deflection plane **414** to be multiplexed into the desired target output ports **402** when the first and second programmable deflection planes are coplanar. Specifically, with one or three separated optical elements in the optical path between the first and second programmable deflection planes, the spectra imaged on the second programmable deflection plane will be inverted (i.e. back to front) compared to the spectra on the first programmable deflection plane. In other words, moving from one side of the first programmable deflection plane to the other the spectra are dispersed from high to low wavelengths, whereas moving in the same direction from one side of the second programmable deflection plane to the other the spectra are dispersed from low to high wavelengths.

[0098] This issue applies unless the first and second programmable deflection planes are facing each other. In the example of FIG. **4**, the first and second programmable deflection planes may be incorporated on the same planar SLM device, thus the first and second programmable deflection planes are coplanar. This issue applies to programmable deflection planes on the same SLM device, and also programmable deflection planes on separate but coplanar SLM devices. It also applies to SLM devices which require the light to be retroreflected. The issue applies more generally where the incidence normal of the first programmable deflection plane is within the same hemispherical angular area as the incidence normal of the second programmable deflection plane. FIG. **5** illustrates an example in which a first programmable deflection plane **501** with incidence normal **502** is at an angle to a second programmable deflection plane **503** with incidence normal **504**. Although not coplanar, the incidence normal **502** and **504** are within the same hemispherical angular area.

[0099] FIG. **5** illustrates the incidence normals **502** and **504** of the physical planes in which the deflection planes are positioned. Where local optics are used to project the location of the deflection plane onto another virtual plane, then the incidence normal is that of the virtual plane. For example, FIG. **6** illustrates the direction of the incidence normal **601** of each of four programmable deflection planes **602***a*, **602***b*, **602***c*, **602***d* which are relocated virtually by means of prism **603**. For packaging purposes, SLM planes may usefully be located on a base plate **604** of an optical switch. Local optics, such as prism(s) or mirror(s) are then used to relocate the programmable deflection planes to virtual locations in the optical switch as desired, in the case of FIG. **6**, plane **605**.

[0100] Utilising an odd number of at least five optical elements in a Fourier configuration in the beam steering optical element group between the two programmable deflection planes ensures that a non-inverted image in the dispersion plane and Fourier conjugate orthogonal to the dispersion plane image of the first programmable deflection plane is formed on the second programmable deflection plane when the incidence normal of the first and second programmable deflection planes are within the same hemispherical angular area.

[0101] Alternatively, an odd number of fewer than five (such as three) optical elements in the beam steering optical element group may be used in combination with diffraction gratings which are of opposing grating order. FIG. **7** illustrates such an example. The features of FIG. **7** which are the same as those of FIG. **4** and operate in the same manner are illustrated with the same reference numerals. The detail of those features is as described above with respect to FIG. **4**. The structure and operation of the M×N WSS of FIG. **7** differs from the description of the M×N WSS of FIG. **4** above by the following details.

[0102] As with FIG. **4**, the beam steering optical element group **709** transfers the deflected array of beams from the first programmable deflection plane **408** onto the second programmable deflection plane **414**. However, the beam steering optical element group **709** of FIG. **7** comprises three optical elements: lenses **710** and **715** and beam steering optical device **712**. Lens **710** diverts the deflected array of beams from the first programmable deflection plane **408** towards the beam steering optical device **712**. The lens **710** images the deflected spectra **417** onto the beam steering optical device **712** in the dispersion plane. Lens **710** is positioned such that the spectra **417** and beam steering optical device **712** are in Fourier planes. Suitably, lens **710** only has optical power in the dispersion plane. Suitably, orthogonal to the dispersion plane, lens **710** has no optical power.

[0103] The beam steering optical device **712** redirects the diverted deflected array of beams towards the second programmable deflection plane **414**. Suitably, the beam steering optical device **712** has no optical power in the dispersion plane. Suitably, the beam steering optical device **712** only has power orthogonal to the dispersion plane. Suitably, the beam steering optical device **712** is in a Fourier configuration with the first and second programmable deflection planes orthogonal to the dispersion axis. In other words, the focal length of the beam steering optical device **712** is equal to the distance between the beam steering optical device **712** and the first and second programmable deflection planes. The beam steering optical device **712** may be, for example, a curved mirror.

[0104] Lens **715** diverts the redirected array of beams from the beam steering optical device **712** towards the second programmable deflection plane. Suitably, lens **715** only has optical power in the dispersion plane. Suitably, orthogonal to the dispersion plane, lens **715** has no optical power.

[0105] Each of the optical elements in the beam steering optical element group **709** has optical power.

[0106] The beam steering optical element group **709** forms an inverted image of the first programmable deflection plane on the second programmable deflection plane. The beam steering optical element group **709** forms a Fourier conjugate image of the first programmable deflection plane on the second programmable deflection plane in a direction orthogonal to the dispersion plane.

[0107] Diffraction grating **701** demultiplexes the signal from the input ports **401** as described with respect to FIG. **4**. Diffraction grating **702** multiplexes the spectra deflected from the second programmable deflection plane **414** to the output ports **402** as described with respect to FIG. **4**. However, the dispersion direction in the dispersion plane of the diffraction grating **701** is of the opposite sign to the dispersion direction in the dispersion plane of the diffraction grating **702**. This enables the inverted spectra on the second programmable deflection plane to be inverted again as they are combined at diffraction grating **702** such that the resulting combined signals are correctly matched to the output ports **402**. For example, diffraction grating **701** may have order +1 and diffraction grating **702** may have order −1. Alternatively, diffraction grating **701** may have order −1 and diffraction grating **702** may have order +1. Alternatively, diffraction grating **701** may have order +2 and diffraction grating **702** may have order −2. Alternatively, diffraction grating **701** may have order −2 and diffraction grating **702** may have order +2. Suitably, the diffraction gratings **701** and **702** have the same magnitude of order, for example both 1 or both 2. Alternatively, the diffraction gratings **701** and **702** may have different magnitudes of order, for example one 1 and the other 2. Since the diffraction gratings **701** and **702** have opposing signs, they are different diffraction gratings. In other words, the same diffraction grating is not used for both diffraction grating **701** and diffraction grating **702**. Utilising two diffraction gratings along with any supporting optics increases the size and weight of the optical switch and the manufacturing cost. Additionally, tolerance issues may occur due to differences in the operation of the diffraction gratings.

[0108] FIGS. **4** and **7** illustrate two different examples of optical switches which optically route light from a set of input ports to a set of output ports via two programmable deflection planes. In order to enable this optical routing to occur effectively, the dispersion direction in the dispersion plane of spectra formed on the second programmable deflection plane with respect to the diffraction order of the diffraction grating which multiplexes the spectra from the second programmable deflection plane is such that the multiplexed signals match the set of output ports. The dispersion direction in the dispersion plane of spectra formed on the second programmable deflection plane is determined by the constituent parts of the beam steering optical element group.

[0109] FIG. **8** illustrates the detailed optical path of an exemplary beam steering optical element group comprising five optical elements. The upper part of FIG. **8** illustrates the optical path in the switching plane, i.e. the y axis shown is the switching axis and the z axis is the optical axis. The lower part of FIG. **8** illustrates the optical path in the dispersion plane, i.e. the x axis shown is the dispersion axis and the z axis is the optical axis. The features of FIG. **8** which are the same as those of FIG. **4** and operate in the same manner are illustrated with the same reference numerals.

[0110] As discussed with reference to FIG. **4**, the beam steering optical element group redirects light from the first programmable deflection plane **408** to the second programmable deflection plane **414**. Four of the five optical elements of the beam steering optical element group have optical power in the dispersion axis. Specifically, lenses **410**, **411**, **413** and **415** have optical power in the dispersion axis. Suitably, lenses **410**, **411**, **413** and **415** have optical power only in the dispersion axis. Beam steering optical device **412** is a lens with optical power in the switching axis. Suitably, beam steering optical device **412** has optical power only in the switching axis. Lens **410** is located one focal length from the first programmable deflection plane **408**. Lens **411** is located two focal lengths from lens **410** and one focal length from beam steering optical device **412**. Beam steering optical device **412** is located one focal length from lens **411** and one focal length from lens **413**. Lens **413** is located one focal length from beam steering optical device **412** and two focal lengths from lens **415**. Lens **415** is located one focal length from lens **413** and one focal length from the second programmable deflection plane **414**. The total optical length from the first programmable deflection plane to the second programmable deflection plane is 8 focal lengths (8f).

[0111] FIG. **9** illustrates an exemplary add-drop WSS **900** in which a single panel **907** comprises two programmable deflection planes **908**, **914**. This single panel may, for example, be an SLM device.

[0112] In the adWSS of FIG. **9**, light from a set of N ports **901** is routed to a set of K drop ports **902**. Each port may comprise one or more optical fibres. Each port transports an optical signal having at least one component data channel. Typically, each port has a plurality of data channels. Each channel transports data. Each channel has a frequency band which has a different centre frequency to the other channels of that port. The channels of a port may have the same or different bandwidths. The bandwidths of the channels of a port are non-overlapping. Each channel from each input port **901** may be routed to any individual output port **902**.

[0113] The optical signals from the input ports **901** pass through coupling lenses **903** which form an optical beam of the N ports before a 4F imaging system. There is a coupling lens for each input port. The 4F imaging system comprises two lenses **904** and **906** separated by a diffraction grating **905***a* at the Fourier plane of lens **904**. The diffraction grating **905***a* spatially separates the component frequency channels of the optical signal beams incident on it to form a set of dispersed optical signal beams. In other words, each optical signal from an input port is spread into its frequency spectrum by the diffraction grating **905***a*. Thus, the diffraction grating **905***a* acts to demultiplex each optical signal into its component data channels.

[0114] A first programmable deflection plane **908** is positioned in the optical path of the set of dispersed optical signals such that the set of dispersed optical signal beams form spectra **917** on the first programmable deflection plane **908**. In the example of FIG. **9**, three input ports **901** are illustrated, each of which forms a spectrum **917** on the first programmable deflection plane **908**. Each spectrum forms an array of beams of the component data channels of the input port which are imaged on the first programmable deflection plane **908**, each beam being a data channel. The first programmable deflection plane **908** applies a programmable deflection to each beam of the array of beams individually to form a deflected array of beams. The degree of deflection applied to each beam is so as to change the angle of that beam to direct its data channel on a path towards a selected one of the output ports **902**. The deflection applied by the first programmable deflection plane **908** is reconfigurable. Thus, a controller providing a control signal to the first programmable deflection plane **908** may change the deflection applied by the first programmable deflection plane **908** to each individual beam of the spectra from each input port, so as to output the desired channels to the desired ports.

[0115] The deflected array of beams from the first programmable deflection plane **908** then travels through a lens **909** and diffraction grating **905***b* in the Fourier plane of lens **909**. The diffraction grating **905***b* may be the same as or different to the diffraction grating **905***a*. The diffraction grating **905***b* recombines the deflected spectra for each output port **902**. Thus, the diffraction grating **905***b* acts to multiplex the spectra. An imaging lens **910** images the optical signal from the diffraction grating **905***b* onto a beam steering optical device **911**.

[0116] Beam steering optical device **911** is suitably a retroreflecting optic which redirects the optical signal from the imaging lens **910** towards the second programmable deflection plane **914**.

[0117] The redirected array of beams is then directed through two lenses **912** and **913**. Lenses **912** and **913** are in a telescope arrangement, and act to image the light output from the beam steering optical device **911** onto the second programmable deflection plane **914**. The telescope arrangement may be so as to magnify the redirected array of beams onto the second programmable deflection plane **914**. Since the spectra have been recombined at the diffraction grating **905***b*, the redirected array of beams in FIG. **9** are multiplexed light signals. The second programmable deflection plane **914** acts as a column of vertical space switches **918**, each of which applies a programmable deflection to each beam of the array of beams incident on it. There are the same number of space switches as there are drop ports, in this case K space switches and K drop ports. The deflection corrects for the angle each input port spectrum makes with the space switch so as to efficiently couple each multiplexed light signal to the selected output port. This ensures a directionless operation. The deflection applied by the second programmable deflection plane **914** is reconfigurable. Thus, a controller providing a control signal to the second programmable deflection plane **914** may change the deflection applied by the second programmable deflection plane **914** to each redirected beam from the beam steering optical device **911** so as to output the desired channels to the desired ports.

[0118] The light deflected from the second programmable deflection plane **914** passes through lens **915** creating a single beam at the lens array **916** for coupling to the output ports **902**.

[0119] The number of angular switching positions in the spectrum channels **917** is equal to the number of output ports K. The number of switching positions in the space switches **918** is equal to the number of input ports N.

[0120] Where there are N input ports and K output ports, and K>N, N output ports may be used for direct transit from the N input ports. K-N output ports may then be used as drop ports.

[0121] FIG. **10** illustrates an example in which two adWSS structures as described above are incorporated into a single switch as an implementation of a two layer edge reconfigurable optical add-drop multiplexer (ROADM) wavelength selective switch. The adWSS structures illustrated in FIG. **10** are the same as that described with respect to FIG. **9**. However, it is understood that any two of the adWSS architectures described herein may instead be incorporated into the edge ROADM switch. One of the adWSS structures in FIG. **10** is operated in the forward direction described with respect to FIG. **9**. This adWSS structure has transit input ports **1001** through which light is directed to form spectra **917***a* on a first programmable deflection plane of panel **907**, which are then routed via beam steering optical device **911** to space switches **1103***a* on a second programmable deflection plane of panel **907**, following which the light is output via drop ports **1002**. The other adWSS structure in FIG. **10** is operated in the reverse direction to that described with respect to FIG. **9**. This reverse adWSS structure has add ports **1003** through which light is directed to space switches **1103***b* on a third programmable deflection plane of panel **907**, which are then routed via another beam steering optical device **911'** to spectra **917***b* on a fourth programmable deflection plane of panel **907**, following which the light is output via transit output ports **1004**. The other optical components of the two adWSS structures are as described with respect to FIG. **9**.

[0122] Suitably, a single SLM panel **907** incorporates the first, second, third and fourth programmable deflection planes. That single SLM panel is planar. Thus, the incidence normal of each of the first, second, third and fourth programmable deflection planes is in the same direction. The two beam steering optical devices **911** and **911'** are suitably retroreflecting optics.

[0123] FIG. **10** illustrates the ROADM switch using only a single SLM panel **907**. Alternatively, the individual adWSS structures may use separate SLM panels. Each adWSS structure may incorporate both its programmable deflection planes on a single SLM panel. Alternatively, the two programmable deflection planes of each adWSS may be on separate SLM panels. One adWSS may incorporate both its programmable deflection planes on a single SLM panel, whilst the other adWSS may have separate SLM panels for each of its programmable deflection planes. If separate SLM panels are used, the first, second, third and fourth programmable deflection planes may not all be coplanar with each other. However, suitably, all of the programmable deflection planes have incidence normals which are within the same hemispherical angular area. As described with reference to FIG. **5**, the programmable deflection planes may be at an angle to each other. As described with reference to FIG. **6**, local optics may be used to project the location of any one or more deflection plane onto another virtual plane. In this case, the incidence normals to the virtually relocated deflection planes are all within the same hemispherical angular area.

[0124] In addition to the two adWSS structures, the edge ROADM switch of FIG. **10** includes a transit section **1005** which provides an optical path from the first programmable deflection plane to the fourth programmable deflection plane so as to transport an optical signal from a transit input port **1001** to a transit output port **1004**. This transit section is an N×N switch which switches from the N spectra **917***a* on the first programmable deflection plane to the N spectra **917***b* on the fourth programmable deflection plane.

[0125] The transit section **1005** may comprise any of the beam steering optical element groups described herein. The transit section **1005** may comprise a lens and a retroreflecting optic. That retroreflecting optic may be a mirror. The lens and mirror may be used to switch channels between the input spectrum **917***a* and the output spectrum **917***b* in the conjugate plane of the lens. In this example, the spectral direction will be reversed due to lateral inversion in a single lens. This can be countered using diffraction grating **905***a* to reverse the spectral direction relative to diffraction grating **905***a'*.

[0126] FIG. **11** illustrates the optical path of another exemplary transit section of a ROADM switch. The top two sections of FIG. **11** illustrate the optical path in the switching plane. The bottom section of FIG. **11** illustrates the optical path in the dispersion plane. The transit section comprises a lens assembly **1105** comprising two lenses **1105***a* and **1105***b* positioned in the optical path between lenses **906** of the first adWSS and **906'** of the second adWSS. Lenses **1105***a* and **1105***b* are spaced two focal lengths of lens **906** from lens **906**. Lenses **1105***a* and **1105***b* are spaced two focal lengths of lens **906'** from lens **906'**. Lens **906** is spaced one focal length of lens **906** from spectra **917***a* on the first programmable deflection plane. Lens **906'** is spaced one focal length of lens **906'** from spectra **917***b* on the fourth programmable deflection plane. Lens **1105***a* is a normal spherical lens. Lens **1105***b* is a cylindrical lens with power only in its dispersion axis. Lens **1105***a* may be implemented as a curved mirror in order to reflect light to the output spectral plane **917***b*, as will be described below in more detail with reference to FIG. **12**.

[0127] The transit section of FIG. **11** also comprises two further lenses **1101** and **1102**. Lens **1101** is positioned in the optical path between lenses **1105***a*, **1105***b* and lens **906**. Lens **1101** is one focal length from lens **1105***b*, **1105***a* and one focal length from lens **906**. Lens **1102** is positioned in the optical path between lenses **1105***a*,**1105***b* and lens **906'**. Lens **1102** is one focal length from lens **1105***b*, **1105***a* and one focal length from lens **906'**. Lenses **1101** and **1102** are used to prevent lateral inversion of the spectra. Lenses **1101** and **1102** are cylindrical lenses with power only in the dispersion plane.

[0128] FIGS. **12***a* and **12***b* illustrate the optical path of another exemplary transit section. FIG. **12***a* illustrates the optical path in the switching plane, and FIG. **12***b* illustrates the optical path in the dispersion plane. The spectra **917***a* on the first programmable deflection plane are deflected through lens **909** towards diffraction grating **905***b*. A first mirror segment **1201** on the diffraction grating **905***b* is aligned with the focus points close to the axis. This first mirror segment **1201** diverts the deflected spectra which are incident upon it through cylindrical lens **1203** to curved mirror **1105***a*. Only some of the deflected spectra from the first programmable deflection plane are incident upon the first mirror segment **1201**. The deflected spectra which are incident upon the first mirror segment **1201** are those which are designated to pass through the transit section. The first mirror segment **1201** may be a separate component to the diffraction grating **905***b*. Alternatively, the first mirror segment **1201** may be integral with the diffraction grating **905***b*. Cylindrical lens **1203** replaces lens **1105***b* in the example of FIG. **11**. Cylindrical lens **1203** only has power in the dispersion axis. Cylindrical lens **1203** has half the power of lens **1105***b* since the light passes through it twice.

[0129] The deflected spectra then propagate to a second mirror segment **1202** on the diffraction grating **905***b'*. Second mirror segment **1202** is aligned with the focus points close to the axis. This second mirror segment **1202** diverts the deflected spectra incident upon it through lens **909'** to the output spectra **917***b* on the fourth programmable deflection plane on SLM panel **907**. The deflected spectra are merged with the spectra generated by the light from the add ports **1003** and propagated to the transit output ports **1004**.

[0130] The first and second mirror segments **1201** and **1202** may be the same but reversed and flat in the switching axis.

[0131] Light **1204** that is focussed away from the segment **1201** hits the diffraction grating **905***b* where it is recombined and propagates through to lens **910** and on to the drop ports **1002** as previously described.

[0132] In the ROADM of FIG. **10**, suitably the diffraction grating **905***a* and the diffraction grating **905***b* have the same order. Suitably, the dispersion direction in the dispersion plane of the diffraction grating **905***b* is the same sign as the dispersion direction in the dispersion plane of the diffraction grating **905***a*. The diffraction grating **905***b* may be the same as the diffraction grating **905***a*. The diffraction grating **905***b* may be different to the diffraction grating **905***a*.

[0133] In the ROADM of FIG. **10**, suitably the diffraction grating **905***a*' and the diffraction grating **905***b*' have the same order. Suitably, the dispersion direction in the dispersion plane of the diffraction grating **905***b*' is the same sign as the dispersion direction in the dispersion plane of the diffraction grating **905***a*'. The diffraction grating **905***b*' may be the same as the diffraction grating **905***a*'. The diffraction grating **905***b*' may be different to the diffraction grating **905***a*'.

[0134] FIG. **16** illustrates a further example optical switch **1600** in which a transit section and a single adWSS structure are incorporated into a single switch. The adWSS structure shown in FIG. **16** is used to drop ports. However, it will be appreciated that the transit section may instead be incorporated into a single switch with an adWSS structure which is used to add ports. In either case, there are only three programmable deflection planes. The operation of the transit section is as described for any of the transit sections above. The operation of the adWSS structure is as described for any of the adWSS structures above. The features of FIG. **16** which are the same as those of FIG. **10** and operate in the same manner are illustrated with the same reference numerals. The detail of those features is as described above with respect to FIGS. **10** to **12**.

[0135] Suitably, a single SLM panel **907** incorporates all three programmable deflection planes of FIG. **16**. That single SLM panel is planar. Thus, the incidence normal of each of the three programmable deflection planes is in the same direction. The beam steering optical device **911** is suitably a retroreflecting optic.

[0136] FIG. **16** illustrates the optical switch **1600** using only a single SLM panel **907**. Alternatively, two or three separate SLM panels may be used. If separate SLM panels are used, those SLM panels may not all be coplanar with each other. However, suitably, all of the programmable deflection planes have incidence normals which are within the same hemispherical angular area. As described with reference to FIG. **5**, the programmable deflection planes may be at an angle to each other. As described with reference to FIG. **6**, local optics may be used to project the location of any one or more deflection plane onto another virtual plane. In this case, the incidence normals to the virtually relocated deflection planes are all within the same hemispherical angular area.

[0137] FIG. **13** illustrates a further ROADM. The ROADM of FIG. **13** is very similar to that of FIG. **10**. It differs in the configuration of the add ports and the drop ports. Add ports **1303** and drop ports **1302** are both arranged in a grid. The ROADM of FIG. **13** also differs from that of FIG. **10** in that the beam steering optical devices **911** and **911'** are remapping optics. Each one remaps the beams incident on it to form a remapped array of beams which it directs to the subsequent programmable deflection plane. In the specific example of FIG. **13**, the beam steering optical device **911** remaps the column array of beams incident on it to a grid array of beams for output to the second programmable deflection plane, where the space switches on the second programmable deflection plane have a corresponding grid array. In the specific example of FIG. **13**, the beam steering optical device **911'** remaps the grid array of beams incident on it to a column array of beams for output to the fourth programmable deflection plane, where the spectra on the fourth programmable deflection plane have a corresponding column array. Each beam steering optical device reverses the direction of the light of the beam incident on it. It also shifts the beam path incident on it so that the alignment of the beam path with the space switches or spectra on the subsequent programmable deflection plane is optimised. It does this by independently controlling the spatial positioning and/or orientation of each beam from the deflected array of beams from the first programmable deflection plane in the remapped array of beams. In this way, it changes the spatial positioning and/or orientation of at least one beam from the deflected array of beams differently to at least one other beam of the deflected array of beams. The optical device may be a mirror, mirror array, freeform optic or a retroreflecting prism.

[0138] FIG. **14** illustrates an implementation of the beam steering optical device **911**. The beam steering optical device remaps an array of points incident on it into a different conformal pattern. That different conformal pattern matches the geometrical distribution of the space switches on the second programmable deflection plane and the geometrical distribution of the image of the set of output ports on the second programmable deflection plane. The beam steering optical device of FIG. **14** is a prism array **1400**. The prism array **1400** remaps a single column array of beams to a multiple column array of beams, as shown in FIG. **13**. Referring to FIG. **13**, the single column of spectra **917***a* are deflected by the first programmable deflection plane **908** and combined by the diffraction grating **905***b*. They are then imaged by lens **910** to form a set of column points **1401** at the entrance to the prism **1400**. Microlenses may optionally be used to collimate the light at the entrance of the prism **1400**. The prism array **1400** remaps the column points **1401** to a grid of points **1403**.

[0139] The prism array **1400** splits the column points **1401** into groups. FIG. **14** illustrates three groups **1401***a*, **1401***b*, **1401***c*, however it will be understood that more or fewer groups may be used. Group **1401***c* is underneath group **1401***b* which is underneath group **1401***a* in the single column of points **1401**. Each group of points **1401***a*, **1401***b*, **1401***c* is directed to a different retroreflecting structure. Group **1401***a* is incident on structure **1402***a*, and exits structure **1402***a* as group **1403***a*. Group **1401***b* is incident on structure **1402***b*, and exits structure **1402***b* as group **1403***b*. Group **1401***c* is incident on structure **1402***c*, and exits structure **1402***c* as group **1403***c*. Prism structure **1402***a* causes group **1403***a* to be displaced to the side relative to group **1401***a*. Prism structure **1402***b* incorporates a periscope prism transmission which causes group **1403***b* to be displaced vertically relative to group **1401***b*. Prism structure **1402***b* also causes group **1403***b* to be displaced to the side relative to group **1401***b*. Thus, group **1403***b* is a different column of points to group **1403***a* which is positioned next to group **1403***a*. Prism structure **1402***c* causes group **1403***c* to be displaced to the side relative to group **1403***c*. Prism structure **1402***c* incorporates a periscope prism transmission which causes group **1403***c* to be displaced vertically relative to group **1401***c*. Prism structure **1402***c* also causes group **1403***c* to be displaced to the side relative to group **1401***c*. Thus, group **1403***c* is a different column of points to groups **1403***a* and **1403***b* which is positioned next to group **1403***b*. In this way, the single column of points **1401** is remapped to a two-dimensional array of columns **1403**.

[0140] Prism structures **1402**a, **1402**b and **1402**c are non-overlapping and may be part of a single passive prism element.

[0141] If micro lenses are not used, then the optical path through each prism structure for each group is altered such that the total optical path length for each group is the same. This ensures that focussing in the subsequent lens systems **912** and **913** focusses all the beams onto the second programmable deflection plane **914**.

[0142] Returning to FIG. **13**, the transit section **1306** may be as described with reference to FIG. **10**, **11** or **12**. Alternatively, the transit section may also incorporate a remapping optic. An example of a transit section incorporating a remapping optic is shown in FIG. **15**.

[0143] FIG. **15** illustrates the optical path of an exemplary transit section which can be used when there is spectral remapping. The transit section of FIG. **15** enables full access to any channel in any transit port from any transit port without the need for dispersion plane switching on the SLM panel. FIG. **15** illustrates the transit section in the dispersion plane. This figure only illustrates the input to transit section of the architecture. The transit section to output is the same as described with reference to FIGS. **12**a and **12**b.

[0144] The spectra **917**a on the first programmable deflection plane are deflected through lens **909** towards diffraction grating **905**b. A first mirror segment **1501** on the diffraction grating **905**b is aligned with the focus points close to the axis. This first mirror segment **1501** is a flat reflector. This first mirror segment **1501** reflects the deflected spectra which are incident upon it to curved mirror **1502**. The curved mirror **1502** is the length of its focal length from the first mirror segment **1501**. The curved mirror reflects the deflected spectra incident on it to a beam steering optical device **911**. The curved mirror **1502** is the length of its focal length from the beam steering optical device **911**.

[0145] The beam steering optical device **911** remaps the spectra from one geometrical distribution to another as described herein. Specifically, the beam steering optical device **911** orders the spectra into a single vertical column aligned by wavelength. This column is then transmitted to a second spherical mirror **1503**. Spherical mirror **1503** focusses the light onto the same second mirror segment **1202** described with respect to FIGS. **12**a and **12**b. The remainder of the transit section is as described with respect to FIGS. **12**a and **12**b.

[0146] Light that is focussed away from the segment **1501** hits the diffraction grating **905**b where it is recombined and propagates through to lens **910** and on to the drop ports **1302** as previously described.

[0147] In the transit sections described herein, by avoiding passing the light through the diffraction gratings between the first and fourth programmable deflection planes, contentionless transit switching is achieved and insertion loss is minimised.

[0148] The beam steering optical device **911** utilised in the example of FIG. **15** between the input and output spectral planes on the SLM panel **907** enables the edge ROADM switch to be a full N×N WSS for the direct transit path from the input ports **1001** to the output ports **1004**. The add ports **1303** and drop ports **1302** are used with the space switch elements as described above for an adWSS in order to add and drop data from the edge ROADM switch.

[0149] FIG. **17** illustrates a further M×N WSS **1700** between input ports **1701** and output ports **1702**. Input light beams from N input ports **1701** pass through a fan lens **1703**, a mirror optic **1704** and an optical system **1705** before being incident on a first programmable deflection plane **1706**. The mirror optic **1704** may be a split aperture mirror (SAM). The SAM may comprise a pair of mirrors separated by a gap. The fan lens **1703** focuses the light input by the input ports **1701** through the gap between the pair of mirrors of the SAM **1704** towards the optical system **1705**. In other words, the fan lens focuses the light to a point aligned with the aperture of the SAM. The input light beams then pass through the optical system **1705** towards the first programmable deflection plane **1706**.

[0150] The optical system **1705** comprises a 4F imaging system. The optical system **1705** therefore images the optical signal in the dispersion plane using two lenses which have optical power in the dispersion direction. The optical system **1705** seen in FIG. **17** also includes a demultiplexer. The demultiplexer is located in the Fourier plane between the two lenses having optical power in the dispersion direction. The demultiplexer is located at the Fourier plane between the two lenses. The demultiplexer in optical system **1705** spreads and collimates the incoming light in the dispersion plane onto the first programmable deflection plane **1706**. The incoming light therefore forms spectra on the first programmable deflection plane **1706**. The optical system **1705** further includes a lens with optical power in the steering direction located between the two lenses which have optical power in the dispersion direction. In this system, the steering direction is orthogonal to the dispersion direction. The lens may be a Fourier lens. The lens having optical power in the steering direction may be in a Fourier configuration with the first programmable deflection plane **1706** and the mirror optic **1704**. The optical system **1705** thus additionally forms a Fourier conjugate in a direction normal to the beam path, along the steering axis. In other words, the optical system **1705** also creates a conjugate Fourier plane in the steering direction. As described, the optical system **1705** may include only a single lens having optical power in the steering direction. Alternatively, the optical system **1705** may take the form seen in FIG. **18**, which includes three lenses having optical power in the steering direction, as described in more detail below.

[0151] According to further examples, the optical system **1705** may have any odd number of lenses having optical power in the steering direction.

[0152] The demultiplexed light incident on the first programmable deflection plane **1706** is deflected in the steering direction and passed back through the optical system **1705** to the mirror optic **1704**. Since the demultiplexed light passes back through optical system **1705** in the opposite direction, the demultiplexer acts as a multiplexer in the reverse direction. The demultiplexed light incident on the optical system **1705** is therefore recombined as it passes back through the optical system. Due to the angular deflection imparted on the light by the first programmable deflection plane, once the light has been deflected by the programmable deflection plane **1706** and has passed back through optical system **1705**, the deflected multiplexed light is intercepted by one of the mirrors in the pair of mirrors in the SAM **1704**. The mirror optic **1704** thus deflects the light in a direction away from input ports **1701**.

[0153] The deflected light then passes through optical system **1707** and is incident on an optical structure **1708**. Optical system **1707** comprises a 4F imaging system. The optical system **1707** comprises two lenses which have optical power in the dispersion direction and image the optical signal in the dispersion plane. The optical system additionally forms a Fourier conjugate in a direction normal to the beam path, along the steering axis. For example, the optical system **1707** further includes a lens having optical power in the steering direction located between the two lenses having optical power in the dispersion direction. The lens may be a Fourier lens. The lens having optical power in the steering direction may be in a Fourier configuration with the second programmable deflection plane **1710** and the optical structure **1708**.

[0154] The optical structure **1708** is configured to alter the configuration of the beams incident on it so as to generate a gap between those beams. Optical structure **1708** may be known as a gap optic. The optical structure **1708** may comprise a mirror array positioned such that light beams incident on it are split into a first set of beams and a second set of beams, where the first and second sets of beams exit the structure as two distinct spatially separated groups of beams. In other words, the optical structure **1708** splits the incoming light into two portions of light separated by a gap. The first and second sets of beams output from the optical structure **1708** are parallel and separated from each other by a gap. The mirror optic **1704** previously described may additionally take the form of a gap optic. The light output by the optical structure **1708** is input to optical system **1709**. Optical system **1709** may take the same form as optical system **1705** previously described. For example, optical system **1709** may comprise a set of two lenses having optical power in the dispersion direction and a lens configured to product a Fourier conjugate in the steering direction. Light deflected by the first programmable deflection plane **1706** therefore passes through optical system **1705**, mirror optic **1704**, optical system **1707**, optical structure **1708** and optical structure **1709** before it is incident on the second programmable deflection plane **1710**. The light incident on the second programmable deflection plane **1710** is multiplexed light. The second programmable deflection plane **1710** is therefore a space switch plane which includes space switch elements configured to deflect the incident multiplexed light. The second programmable deflection plane **1710** angularly deflects the incoming light such that it passes through the gap created in the incoming light by optical structure **1709**. The deflected light thus passes through optical system **1709** and the optical structure **1708** without interfering with oncoming light. The deflected light passes through the mirror array of optical structure **1708** undeflected and is distributed to the output ports **1702** by fan lens **1711**.

[0155] According to other examples, the switch illustrated in FIG. **17** may comprise a further demultiplexer, for example as part of optical structure **1709**, such that spectra are produced on both the first programmable deflection plane **1710** and the second programmable deflection plane **1710**. As previously described, the first and second programmable deflection planes **1706**, **1710** may be located on separate SLM devices or may be located on a single common SLM device.

[0156] As previously mentioned, FIG. **18** illustrates an example of the optical system **1705** seen in FIG. **17**. As described, the optical system **1705** is positioned between mirror optic **1704** and the first programmable deflection plane **1706**. The optical system according to this example comprises two lenses in the dispersion plane **1801***a*, **1801***b*. Lenses **1801***a*, **1801***b* may be cylindrical lenses with power in the dispersion plane. The optical system **1705** further comprises three lenses having optical power in the steering plane, **1803***a*, **1803***b*, **1803***c*. The lenses **1803***a*, **1803***b*, **1803***c* may be cylindrical lenses having optical power in the steering direction. Lens **1803***a* is located a distance from mirror optic **1704** equal to the focal length of lens **1803***a*. Steering lenses **1803***b* and **1803***c* are positioned between the two dispersion lenses **1801***a* and **1801***b*. Lenses **1801***a*, **1801***b* may be in a 4F configuration. In other words, lens **1704** is one focal length from lens **1801***a*, lens **1706** is one focal length from lens **1801***b* and lenses **1801***a* and **1801***b* are separated by a distance which is equal to the sum of their focal lengths.

[0157] The optical system further comprises a demultiplexer **1802**. The demultiplexer **1802** may be a diffraction grating. The demultiplexer **1802** is located between the two dispersion lenses **1801***a*, **1801***b*. The demultiplexer **1802** may be positioned in the plane that is one respective focal length from lens **1801***a* and one respective focal length from lens **1801***b*. The demultiplexer **1802** is in a Fourier conjugate plane in the steering direction to the first programmable deflection plane **1706**.

[0158] Steering lenses **1803***a*, **1803***b*, **1803***c* form a Fourier conjugate plane in the steering direction. For example, lenses **1803***b* and **1803***c* may form a 4F imaging system between the first programmable deflection plane **1706** and a plane which is a distance from lens **1803***a* equal to the focal length of lens **1803***a*. The steering lens **1803***c* may be coincident with dispersion lens **1801***b* between the demultiplexer **1802** and the first programmable deflection plane **1706**. Steering lens **1803***b* may be coincident with the demultiplexer **1802**. According to further examples, lenses **1803***c* and **1801***b* are in the same Fourier conjugate relationship with the plane of the demultiplexer **1802** and hence have the same focal length in the steering and dispersion directions. Lenses **1803***c* and **1801***b* are positioned at the same location and have the same focal length. Thus in the simplest arrangement, lenses **1803***c* and **1801***b* may be replaced by a single lens having optical power in the dispersion direction and in the steering direction. Lenses **1801***b* and **1803***c* may be circularly symmetric. Lenses **1801***b* and **1803***c* may be a single spherical lens of the same focal length in both direction.

[0159] FIG. **19** illustrates a further ROADM switch **1900** which is a two-layer edge reconfigurable optical multiplexer (ROADM) wavelength selective switch. The switch **1900** seen in FIG. **19** is a twin system in a single package formed from two of the M×N WSSs **1700** seen in FIG. **17**. The WSSs (**1700***a*, **1700***b*) are arranged on top of one another (in the steering direction). The components of the switch **1700** seen in FIG. **17** and described above are duplicated in the switch **1900** seen in FIG. **19**. The first M×N WSS **1700***a* comprises a first programmable deflection plane **1706***a*. The second M×N WSS **1700***b* comprises a second programmable deflection plane **1706***b*. The first M×N WSS **1700***a* comprises a third programmable deflection plane **1710***b*. The second M×N WSS **1700***b* comprises a fourth programmable deflection plane **1710***a*. In the switch **1900** seen in FIG. **19**, the first and second programmable deflection planes **1706***a*, **1706***b* are incorporated into a single SLM device **1706**. The third and fourth programmable deflection planes **1710***b*, **1710***a* are incorporated into a single SLM device **1710**. In other examples, the first and second programmable deflection planes **1706***a*, **1706***b* may be located on different devices. The third and fourth programmable deflection planes **1710***b*, **1710***a* may be located on different devices. According to further examples, all four programmable deflection planes **1706***a*, **1706***b*, **1710***a*, **1710***b* may be incorporated into a single device. As will be described in more detail in the following examples, in some twin systems, equivalent components in each of the two WSSs are combined into a single component for the twin system. For example, mirror optics **1704***a*, **1704***b* may be integrated into a single mirror optic **1704** for the twin system. Each M×N WSS **1700***a*, **1700***b* is configured to operate as previously described so as to transfer signals from ports **1701***a* to ports **1702***a*, from ports **1701***b* to **1702***b*, from ports **1702***a* to ports **1701***a* and from ports **1702***b* to ports **1701***b*.

[0160] FIG. **20** illustrates a further ROADM switch **2000**. The structure of the switch **2000** is a two-level system formed of M×N WSSs **1700***a* and **1700***b* having the components seen in FIG. **19** and described above. Each M×N WSS **1700***a*, **1700***b* is configured to operate as previously described so as to transfer signals from ports **1701***a* to ports **1702***a*, from ports **1701***b* to **1702***b*, from ports **1702***a* to ports **1701***a* and from ports **1702***b* to ports **1701***b*.

[0161] Similarly to the switches seen in FIGS. **10**, **13** and **16**, the switch **2000** additionally includes a transit section **2001**, also referred to as a beam steering optical device, which enables signals to be transferred from one of systems **1700***a*, **1700***b* to the other of systems **1700***a*, **1700***b*. This transit section is an N×N switch which switches from N spectra on the first programmable deflection plane **1706***a* of the first WSS **1700***a* to N spectra on the second programmable deflection plane **1706***b* of the second WSS **1700***b*. The transit section can be positioned at any plane that contains a conjugate Fourier image of the spectral plane of the any of the programmable deflection planes focused in the steering direction. In the switch **2000**, the transit section **2001** is positioned adjacent to the demultiplexer **1802** of optical systems **1705***a*, **1705***b*. The transit section **2001** enables signals to be transferred from ports **1701***a* (acting as input ports) to ports **1701***b* (acting as output ports) as indicated by arrows in FIG. **20**, or from ports **1701***b* (acting as input ports) to ports **1701***a* (acting as output ports).

[0162] According to one example, the optical path of a signal input from input ports **1701***a* through elements of the WSS **1700***a* is as follows. The signal passes through fan lens **1703***a* and through a gap in SAM **1704***a* to optical system **1705***a*. The input light beams then pass through the optical system **1705***a* towards the first programmable deflection plane **1706***a*. The optical system **1705***a* in switch **2000** takes the form seen in FIG. **18**. The optical system **1705***a* also includes a demultiplexer **1802***a*. As previously described, the optical system **1705***a* images the optical signal in the dispersion plane using two lenses in a 4F configuration having optical power in the dispersion direction and forms a Fourier conjugate image in the steering direction at the demultiplexer. The demultiplexer in optical system **1705***a* spreads and collimates the incoming light in the dispersion plane onto the first programmable deflection plane **1706***a*. As previously explained, the demultiplexer in optical system is located between the two 4F dispersion lenses. As described above, in the optical system **1705***a* of the type seen in FIG. **18**, lenses **1803***c* and **1801***b* may be part of a single spherical lens such that the optical system comprises four lenses and one demultiplexer. The incoming light therefore passes through one lens having optical power in the dispersion direction (**1801***a*), two lenses having optical power in the steering direction (**1803***a*, **1803***b*), a lens having optical power in both directions (**1803***c*+**1801***b*) and a demultiplexer (**1802**) before forming spectra on the first programmable deflection plane **1706***a*. The incoming light passes through a gap **2102***a* in the demultiplexer (described below) where the light is demultiplexed. The demultiplexed light incident on the first programmable deflection plane **1706***a* is deflected in the steering direction and is incident again on the spherical lens (**1803***c*+**1801***b*) then the demultiplexer. Specifically, each individual frequency band of the demultiplexed light is deflected in the steering direction by an amount related to the position of the required output ports **1701***b*.

[0163] As previously explained, each M×N WSS **1700***a*, **1700***b* includes an optical system **1705***a*, **1705***b*, each comprising a demultiplexer **1802***a*, **1802***b* positioned between pairs of 4F dispersion lenses. In the present example, demultiplexers **1802***a*, **1802***b* are integrated into a common device **1802**. According to other examples, the demultiplexers **1802***a*, **1802***b* may be present on separate devices.

[0164] FIG. **21** illustrates the demultiplexer **1802** in optical system **1705***a* and the transit section **2001** in more detail. FIG. **21** illustrates that the demultiplexer **1802** includes a plurality of ports. As previously described, the majority of ports of the upper set of ports receive light deflected by the first programmable deflection plane **1706***a* and re-multiplex the signal before it is passed through the WSS **1700***a* and propagated through switch **2000** i.e. to mirror optic **1704***a* and onto output ports **1702***a* taking a path as described with respect to FIG. **17**. The same is true of the lower set of ports with respect to the WSS **1700***b* although the signal may be passing through the WSS in the opposite direction (from **1702***b* to **1701***b*). The demultiplexer **1802** includes four transit input ports **2101***a* for sending the received signal to the transit section and four transit output ports **2101***b* for receiving the signal from the transit section. According to other examples, the demultiplexer may comprise more or fewer transit input ports and/or outport ports. As seen in FIG. **21**, the four transit input ports **2101** are positioned as a group of four next to a gap in the ports **2102***a*. The same is true of the four transit output ports **2101***b* and gap **2101***b*. In the majority of examples, the transit input ports and transit output ports are the ports closest to the respective gap, i.e. the lowest loss ports. According to another example, the transit input ports **2101***a* may be divided into two pairs by a central gap and the transit output ports **2101***b* may be divided into two pairs by a central gap. As previously explained, the signal input from input ports **1701***a* passes through the gap **2102***a* in demultiplexer **1802** on route to the first programmable deflection plane **1706***a*. Specifically, it is the case that the incoming signal passes through the gap **2102***a* where it is split into spectra by the demultiplexer and is incident on the first programmable deflection plane **1706***a*. Once the demultiplexed light has been angularly deflected by the first programmable deflection plane **1706***a* and is again incident on the demultiplexer **1802**, a portion of the deflected demultiplexed light is incident on the transit input ports **2101**. The other portion of the light is incident on others of the ports and propagates through the WSS **1700***a* as previously described. For the portion of the light incident at the transit input ports **2101***a*, mirrors located at the transit input ports **2101***a* deflect the incoming light towards the transit section **2001**. The mirrors located at the transit input ports **2101***a* may be at an angle to the surface of the demultiplexer **1802** but this is not required. The mirrors may be flush with the demultiplexer.

[0165] The transit section **2001** may comprise any of the beam steering optical element devices described herein. The transit section **2001** may comprise a lens and a retroreflecting optic. That retroreflecting optic may be a mirror. The transit section **2001** seen in FIG. **21** includes two cylindrical lenses **2103***a*, **2103***b* having optical power in the dispersion direction. The two lenses **2103***a*, **2103***b* may be identical lenses or different lenses. In the example seen in FIG. **21**, the two lenses are identical and are made into a single extended cylindrical dispersion lens **2103** through which both the input and output signal pass. The lens **2103** may be located a distance of a single focal length from the transit input ports **2101***a* and transit output ports **2101***b*. The transit section **2001** further includes a pair of reflecting mirrors **2104***a*, **2104***b*. The first reflecting mirror **2104***a* reflects light from the normal to the mirror at the top transit input port **2101** to the normal to the mirror at the bottom transit output mirror. A lens **2105** having power in the steering direction is positioned between the two reflecting mirrors **2104***a*, **2104***b* in the steering direction. As seen in FIG. **21**, the lens **2105** is mounted horizontally between the mirrors **2104***a* and **2104***b*. The lens **2105** is positioned optically equidistant from the reflecting mirror **2104***a* and from the reflecting mirror **2104***b*. The Fourier conjugate lens **2105** has a focal length equal to the optical distance of the beam path between the lens **2105** and the transit input ports **2101***a* and between the lens **2105** and the transit output ports **2101***b*. The focal length of lens **2105** may be chosen so that the beam waist in the steering direction is the same at the transit input ports **2101***a* as at the transit output ports **2101***b*. The focal length of each of the cylindrical dispersion lenses **2103***a*, **2103***b* may be half the length of the focal length of the Fourier conjugate lens **2105**. The reflecting mirrors **2104***a*, **2104***b* in the example seen in FIG. **21** are orthogonal to one another and rotated relative to one another about axes A and B seen in FIG. **21** which are both normal to the optical axis and parallel to the plane of the demultiplexer. Positioning mirrors **2104***a*, **2104***b* orthogonally with respect to each other introduces a 180 degree rotation of the optical signal. A rotation by this amount is not undesirable as when the signal has a vertical column configuration, the vertical column configuration remains after reflection by mirrors **2104***a*, **2104***b*. According to further examples, the mirrors **2104***a*, **2104***b* are not perfectly orthogonal. This arrangement may allow for more control of the positions of the transit input ports **2101***a* and transit output ports **2101***b*.

[0166] Upon leaving the transit section **2001**, the signal is incident on the transit output ports **2101***b* of demultiplexer **1802**. Mirrors located at the transit output ports **2101***b* deflect the light towards the second programmable deflection plane **1706***b* via spherical lens (**1803***c*+**1801***b*) of optical system **1705***b*. The mirrors located at the transit output ports **2101***b* may be at an angle to the surface of the demultiplexer **1802** but this is not required. The mirrors may be flush with the demultiplexer. The optical system **1705***b* has the form seen in FIG. **18**. The second programmable deflection plane **1706***b* corrects the angle of the incoming signal and deflects the signal back towards the optical system **1705***b*. In other words, the light is directed back through the demultiplexer **1802** which, in the opposite direction, acts as a multiplexer. The deflected signal passes through the gap **2102***b* where it is recombined by the multiplexer **1802**. The re-multiplexed signal then passes though a gap in mirror optic **1704***b* and through fan lens **1703***b* towards the output ports **1701***b*.

[0167] In summary, for ROADM switch **2000**, the optical path between the first programmable deflection plane **1706***a* of the first M×N WSS **1700***a* and the second programmable deflection plane **1706***b* of the second M×N WSS **1700***b* involves passing spherical lens (**1803***c*+**1801***b*) (of optical system **1705***a*), being reflected by a mirror at the transit input ports **2101***a* of demultiplexer **1802**, passing through cylindrical lens **2103***a*, being reflected by mirror **2104***a*, passing through lens **2105**, being reflected by mirror **2104***b*, passing through cylindrical lens **2103***b*, being reflected by a mirror at the transit output ports of **2101***b* of demultiplexer **1802** and passing through spherical lens (**1803***c*+**1801***b*) (of optical system **1705***b*) before being incident on the second programmable deflection plane **1706***b*. For optical signals entering the switch **2000**, the optical path between the first programmable deflection plane **1706***a* of the first M×N WSS **1700***a* and the second programmable deflection plane **1706***b* of the second M×N WSS **1700***b* therefore features five elements having optical power. The five elements in the example seen in FIG. **20** are the two spherical lenses (**1803***c*+**1801***b*) (of **1705***a* and **1705***b*) and the two cylindrical dispersion lenses **2103***a*, **2103***b* and the Fourier conjugate lens **2105** of the transit section **2001**. Light input to the switch via input ports **1701***a* passes twice through the multiplexer **1802** before it is output to ports **1701***b*. In other words, the light is demultiplexed once and re-multiplexed once. Furthermore, as described, demultiplexed light is passed through the transit section **2001**.

[0168] FIG. **22** illustrates a further ROADM switch **2200**. The structure of the switch **2200** is the same as that of switch **2000** previously described (and shown in FIG. **20**) except for the location of the transit section **2201**. In contrast to the switch **2200** in which the transit section **2001** is positioned adjacent to the demultiplexer **1802** of optical systems **1705**, **1705**, in the switch **2200**, the transit section **2201** is located adjacent to the mirror optics **1704***a*, **1704***b*.

[0169] Each M×N WSS **1700***a*, **1700***b* is configured to operate as previously described so as to transfer signals from ports **1701***a* to ports **1702***a*, from ports **1701***b* to **1702***b*, from ports **1702***a* to ports **1701***a* and from ports **1702***b* to ports **1701***b*. In addition, the switch **2200** includes transit section **2201** which enables signals to be transferred from one of systems **1700***a*, **1700***b* to the other of systems **1700***a*, **1700***b*. The transit section **2001** enables signals to be transferred from ports **1701***a* (acting as input ports) to ports **1701***b* (acting as output ports) as indicated by arrows in FIG. **22**, or from ports **1701***b* (acting as input ports) to ports **1701***a* (acting as output ports).

[0170] According to one example, the optical path of a signal input from input ports **1701***a* passes through elements of the WSS **1700***a* as follows. The signal passes through fan lens **1703***a* and through a gap **2302***a* in mirror optic **1704***a* to optical system **1705***a*. As previously described, the mirror optic **1704***a* may comprise a pair of mirrors separated by a gap. The input light beams then pass through the optical system **1705***a* towards the first programmable deflection plane **1706**. As previously described the optical system **1705***a* images the optical signal in the dispersion plane using two lenses in a 4F configuration both having optical power in the dispersion direction. The optical system **1705***a* further includes a lens with optical power in the steering direction located between the two lenses which have optical power in the dispersion direction. The optical system **1705***a* also includes a demultiplexer **1802***a*. The demultiplexer in optical system **1705***a* spreads and collimates the incoming light in the dispersion plane onto the first programmable deflection plane **1706***a*. As previously explained, the demultiplexer in optical system is located between the two lenses having optical power in the dispersion direction. The demultiplexer is located at the Fourier plane between the two lenses. The optical system **1705***a* in switch **2200** requires only one lens having optical power in the steering direction.

[0171] The incoming light therefore passes through two lenses having optical power in the dispersion direction and one lens having optical power in the steering direction before forming spectra on the first programmable deflection plane **1706***a*. The demultiplexed light incident on the first programmable deflection plane **1706***a* is deflected in the steering direction as required for each frequency and passed back through the optical system **1705***a* to the mirror optic **1704***a*. Since the demultiplexed light passes back through optical system **1705***a* in the opposite direction, the demultiplexer acts as a multiplexer in the reverse direction. The demultiplexed light incident on the optical system **1705***a* is therefore recombined as it passes back through the demultiplexer **1802**.

[0172] As previously explained, each M×N WSS **1700***a*, **1700***b* includes a mirror optic **1704***a*, **1704***b*. In the present example, mirror optics **1704***a*, **1704***b* are integrated into a common mirror optic **1704**. According to other examples, the mirror optics **1704***a*, **1704***b* may be present on separate devices.

[0173] FIG. **23** illustrates the mirror optic **1704** and the transit section **2201** of switch **2200** in more detail. The mirror optic **1704** takes the form of a split aperture mirror featuring two gaps **2302***a*, **2302***b*. A first angled mirror is situated adjacent to the first gap **2302***a*. A second angled mirror is situated adjacent to the second gap **2302***b*. The mirror optic **1704** includes four transit input ports **2301***a* and four transit output ports **2301***b*. As seen in FIG. **23**, the four transit input ports **2301***a* are located on the first angled mirror and are located on one side of the first gap **2302***a*. The four transit output ports **2301***b* are located on the second angled mirror and are located on one side of the second gap **2302***b*. According to another example, the transit input ports **2301***a* may be divided into two pairs by a central gap and the transit output ports **2301***b* may be divided into two pairs by a central gap. According to other examples, the mirror optic may comprise fewer or more transit input ports and/or output ports.

[0174] As previously explained, the signal input from input ports **1701***a* passes through the mirror optic **1704***a* on route to the first programmable deflection plane **1706***a*. Specifically, it is the case that the incoming signal passes through the first gap **2302***a* and is passed to optical system **1705***a*. As previously described, the optical system **1705***a* includes a demultiplexer **1802***a* configured to split the incoming multiplexed signal into spectra which are incident on the first programmable deflection plane **1706***a*. In contrast to the arrangement described in FIG. **20**, in the system **2200** seen in FIG. **22**, once the demultiplexed light has been deflected by the first programmable deflection plane **1706***a*, it is passed back through the entirety of optical system **1705***a* and so is re-multiplexed by the demultiplexer **1802***a*, which acts as a multiplexer. The deflected multiplexed light is then passed back to the mirror optic **1704***a*. At this stage, due to the transformation imparted on the deflected signal by the first programmable deflection plane **1706***a*, the deflected light does not pass through the central gap of the mirror optic but instead is intercepted by one of the ports illustrated in FIG. **23**. A portion of the deflected re-multiplexed light is incident of the four transit input ports **2301***a*. The mirror located at the transit input ports **2301***a* deflects the incoming light towards the transit section **2201**.

[0175] The transit section **2201** may take the same form as transit section **2001** seen in FIG. **21**. For example, the transit section **2201** seen in FIG. **23** includes two cylindrical dispersion lenses **2103***a*, **2103***b* made into a single extended cylindrical dispersion lens **2103**, a pair of reflecting mirrors **2104***a*, **2104***b* and a lens **2105** having power in the steering direction positioned between the two reflecting mirrors **2104***a*, **2104***b*.

[0176] In contrast to the arrangement seen in FIG. **20**, in the switch **2200**, multiplexed light is incident on the mirror optic **1704** and the transit section **2201**. The transit input ports **2301***a* and transit output ports **2301***b* therefore take the form of space switch elements. Since the transit section is only required to deal with multiplexed beams with smaller beam sizes, the optics included in the transit section **2201** may be simplified with respect to the transit section **2001**.

[0177] Positioning the transit section adjacent to the mirror optics **1704***a*, **1704***b* as opposed to adjacent to the demultiplexer has a number of advantages. As explained above, the signal passing through the transit section is multiplexed, so the size of each component in the transit section **2201** can be reduced with respect to transit section **2001**. Furthermore, the tolerances of the transit section may be improved as spectra do not need to be dealt with.

[0178] Upon leaving the transit section **2201**, the signal is incident on the transit output ports **2301***b* at the mirror optic **1704**. The second angled mirror located at the transit output port **2301***b* deflects the light towards the second programmable deflection plane **1706***b* of the second M×N WSS **1700***b* through the optical system **1705***b*. The signal therefore passes again through a demultiplexer, this time of the optical system **1705***b* of second M×N WSS **1700***b*. A demultiplexed signal is therefore incident on the second programmable deflection plane **1706***b* which corrects the angle of the incoming signal and deflects the signal back through the optical system **1705***b*. The light is deflected back through the demultiplexer which, in the opposite direction, acts as a multiplexer. At the mirror optic **1704**, the deflected re-multiplexed signal passes through the second gap **2302***b* adjacent the transit output ports **2301***b*. The re-multiplexed signal then passes through fan lens **1703***b* towards the output ports **1701***b*.

[0179] In summary, for ROADM switch **2200**, the optical path between the first programmable deflection plane **1706***a* and the second programmable deflection plane **1706***b* involves passing through two lenses having optical power in the dispersion direction, one lens having optical power in the steering direction and a demultiplexer (as part of optical system **1705***a*), being reflected by a mirror at the transit input ports **2301***a* of mirror optic **1704**, passing through cylindrical lens **2103***a*, being reflected by mirror **2104***a*, passing through lens **2105**, being reflected by mirror **2104***b*, passing through cylindrical lens **2103***b*, being reflected by a mirror at the transit output ports **2301***b* of mirror optic **1704**

and again passing through two lenses having optical power in the dispersion direction, at least one lens having optical power in the steering direction and a demultiplexer (as part of optical system **1705***b*) before being incident on the second programmable deflection plane **1706***b*. For optical signals entering the switch **2200**, the optical path between the first programmable deflection plane **1706***a* of the first M×N WSS **1700***a* and the second programmable deflection plane **1706***b* of the second M×N WSS **1700***b* features nine elements having optical power. The nine elements in the example seen in FIG. **22** are the (cumulative) four lenses having optical power in the dispersion direction and two lenses having optical power in the steering direction of optical systems **1705***a*, **1705***b*, the two cylindrical dispersion lenses **2103***a*, **2103***b* of the transit section **2001** and the lens **2105** of the transit section **2001**. The increase in the number of optical components the signal passes through when compared with the switch **2000** results in increased losses in the optical path between ports **1701***a* and **1701***b*. Light input to the switch **2200** via input ports **1701***a* passes four times through a de-multiplexer **1802***a*, **1802***b* before it is output at ports **1701***b*. In other words, the light is demultiplexed twice and re-multiplexed twice. As described, multiplexed light is passed through the transit section **2201**. In contrast to the two passes through the demultiplexer in switch **2000**, in switch **2200**, the signal passes through the demultiplexer four times which can introduce increased loss due to the higher losses associated with the demultiplexers relative to other optical components in the switch.

[0180] According to one variation of switch **2200**, optical systems **1705***a*, **1705***b* take the form seen in FIG. **18** and previously described as comprising one lens having optical power in the dispersion direction (**1801***a*), two lenses having optical power in the steering direction (**1803***a*, **1803***b*), a lens having optical power in both directions (**1803***c*+**1801***b*) and a demultiplexer (**1802**). According to this example, the optical path between the first programmable deflection plane **1706***a* of the first M×N WSS **1700***a* and the second programmable deflection plane **1706***b* of the second M×N WSS **1700***b* features thirteen elements having optical power.

[0181] According to a further example of a switch in which the transit section is located adjacent the mirror optics **1704***a*, **1704***b*, the combined mirror optic **1704** may take the form of a gap optic configured to alter the configuration of the beams incident on it so as to generate or remove a gap between those beams. FIG. **24** illustrates gap optic **2400** and the path of light beams passing through the gap optic in two opposite directions. As described below, the gap optic **2400** directs incident beams to and from the transit section **2201**.

[0182] According to other examples, each mirror optic **1704***a*, **1704***b* may take the form of a separate gap optic (**2400***a* and **2400***b*). Gap optics **2400***a*, **2400***b* may be separate devices formed of different components or may comprise all or some of the same components.

[0183] Gap optic **2400** comprises a mirror array comprising a first array of mirror surfaces **2402** and a second array of mirror surfaces **2403**. The first array of mirror surfaces **2402** comprises two mirror surfaces **2402***a*, **2402***b* separated by a gap. The first array of mirror surfaces **2402** is spaced apart from the second array of mirror surfaces **2403**. The second array of mirror surfaces comprises two mirror surfaces **2403***a*, **2403***b* with no gap between them.

[0184] Beams incident on the first array of mirror surfaces **2402** from optical system **1705***a* are deflected onto the second array of mirror surfaces **2403** which removes a gap between the beams. Beams which are reflected by the second array of mirror surfaces **2403** are incident on a directing mirror **2404** which directs the beams to the transit section **2201**. Specifically, directing mirror **2404** directs the beams to lens **2103***a* of the transit section. Mirror surfaces **2402**, **2403** and directing mirror **2404** therefore operate together as transit input ports **2401***a* for sending the received signal to the transit section. The beam path through the transit section **2201** is as previously described. Beams from input ports **1701***a* which are incident on the gap **2405** between the two mirror surfaces **2401**, **2401***b*, of the first array of mirror surfaces **2401** pass through the gap optic undeflected and continue towards the first programmable deflection plane **1706***a*.

[0185] Beams which have passed through the transit section **2201** are incident on directing mirror **2404**. The beams are directed to the second array of mirror surfaces **2403**. The second array of mirror surfaces **2403** deflect the beams onto the first array of mirror surfaces **2402** which creates a gap between the beams. The beams are directed by the first array of mirror surfaces **2402** towards optical system **1705***b* and the second programmable deflection plane **1706***b*. Mirror surfaces **2402**, **2403** and directing mirror **2404** therefore operate together as transit output ports **2401***b* for receiving the signal from the transit section. Beams from the second programmable deflection plane **1706***b* which are incident on the gap **2405** between the two mirror surfaces **2401**, **2401***b*, of the first array of mirror surfaces **2401** pass through the gap optic undeflected and continue towards output ports **1701***b*. The gap optic **2400** may alternatively be implemented as part of a switch and used to direct incident beams to and from a transit section located adjacent the demultiplexer of the switch (as seen in FIG. **20**).

[0186] FIG. **25** illustrates an alternative optical arrangement **2502** configured to direct light to and receive light from a transit section. The arrangement may be used in conjunction with a transit section located adjacent the mirror optic **1704** or the demultiplexer **1802** as seen in FIGS. **20** and **22**. The optical arrangement **2502** may therefore be configured to direct and receive multiplexed or demultiplexed light. The arrangement **2502** comprises a first optical group which operates together as transit input ports **2501***a* for directing received signals to the transit section (not shown). The arrangement comprises a second optical group which operates together as transit output ports **2501***b* for receiving the signal from the transit section (not shown).

[0187] The first optical group **2501***a* comprises a polarising beam splitter **2503***a* and a Faraday rotator **2504***a*. The polarising beam splitter is configured to transmit light at a first plane of polarisation and reflect light at a second plane of polarisation which is orthogonal to the first plane of polarisation. The polarising beam splitter **2503***a* is configured to transmit light having a vertical plane of polarisation and reflect light having a horizontal plane of polarisation. Light input from input ports **1701***a* may have a vertical plane of polarisation so that it is transmitted through the polarising beam splitter **2503***a* and then passes through the Faraday rotator **2504***a* on the way to the first programmable deflection plane **1706***a* (not shown in FIG. **25**). As the light passes through the Faraday rotator, the plane of polarisation of the light is rotated by 45 degrees. Light deflected by the first programmable deflection plane **1706***a* is passed back through the Faraday rotator **2504***a* and the plane of polarisation of the light is rotated by another 45 degrees such that the plane of polarisation of the light is now horizontal. Deflected light incident on the polarising beam splitter **2503***a* is therefore reflected by the polarising beam splitter **2503***a* towards the transit section. For example, the polarising beam splitter **2503***a* may direct the light to the lens **2103***a* of transit section **2201**. According to another example, the first optical group may be arranged such that light from the input ports is reflected by the polarising beam splitter on route to the first programmable deflection plane and the light directed to the transit section is transmitted through the polarising beam splitter.

[0188] The second optical group **2501***b* comprises a polarising beam splitter **2503***b*, a Faraday rotator **2504***b* and a half wave plate **2505**. Light which has passed through the transit section is incident on polarising beam splitter **2503***b*. Polarising beam splitter **2503***b* has the same structure as polarising beam splitter **2503**. The light received from the transit section having a horizontal plane of polarisation is therefore reflected by the polarising beam splitter **2503***b* and then passes through the Faraday rotator **2504***b*. The Faraday rotator **2504***b* rotates the plane of polarisation of the light by 45 degrees. The light then passes through half wave plate **2505** which aligns the light with the output ports **1701***b* before the light is incident on the second programmable deflection plane **1706***b*. Light deflected by the second programmable deflection plane **1706***b* is passed back through the half wave plate **2505**, Faraday rotator **2504***b* and the plane of polarisation of the light is rotated by another 45 degrees such that the plane of polarisation is now vertical. Deflected light incident on the polarising beam splitter **2503***b* is therefore transmitted through the polarising beam splitter **2503***b* towards output ports **1701***b* as previously described. Alternatively, light input at ports **1702***b* may not pass through either of the optical groups **2501***a*, **2501***b* on route to the ports **1701***b* such that only light being directed through the transit section interacts with the first and second optical groups **2501***a*, **2501***b*. According to the example seen in FIG. **25**, the two Faraday rotators **2504***a* and **2504***b* are the same Faraday rotator. According to a different example, the two Faraday rotators **2504***a* and **2504***b* are different Faraday rotators, where one of the rotators rotates the beam equally and opposite to the other rotator. In this case, the half wave plate **2505** is not required to be present in optical group **2501***b*.

[0189] As explained with respect to FIGS. **21** and **23**, only some of the light incident on the polarising beam splitter **2503***a* is selected for transit and directed to the transit section. In the examples seen in FIGS. **21** and **23**, the respective demultiplexer **1802** and mirror optic **1704** comprise four transit input ports and four transit output ports. According to other examples, each of the demultiplexer and the mirror optic may comprise fewer or more transit input ports and/or output ports. For example, the demultiplexer and/or the mirror optic may include only one transit input port and one transit output port, as explained in more detail below. According to the arrangement **2502** seen in FIG. **25**, the first optical group **2501***a* may act as only a single transit input port. The second optical group **2501***b* may act as only a single transit output port. For example, only undeviated light **2506***a* may be selected for transit. Undeviated light is light which is deflected by the first programmable deflection plane **1706***a* but has a path through the optical group **2501***a* which is unaffected by the deflection by the first programmable deflection plane. In other words, the path of undeviated light is the same as it passes through the first optical group **2501***a* in both directions. According to this example, only undeviated light **2506***a* is directed to the transit section (and therefore passes to the output ports). This arrangement may be advantageous in situations in which the first programmable deflection plane **1706***a* and/or the second programmable deflection plane **1706***b* fail meaning that light is not deflected by the respective programmable deflection plane and only undeviated light is present in the system. In this scenario, light may be still be passed through the transit section and directed from input ports **1701***a* to output ports **1701***b* meaning that operation of the ROADM switch can continue even after a fault with the first or second programmable deflection planes.

[0190] As discussed, the mirror optic **1704** seen in FIG. **23** may comprise fewer or more transit input ports and/or output ports. According to the example transit schemes illustrated in FIGS. **21** and **23** and previously described, each transit scheme offers a full N×N switch between each of the N input transit ports and the N output transit ports. In the examples seen in FIGS. **21** and **23**, N=4. According to further examples, instead of a single N×N switch, it is possible to use a number P of L×L switches where P=N/L, where N and L are integers. For each L×L switch, only L input transit ports and L output transit ports are used. This reduction in transit input and output ports leads to less switching in transit and reduced loss.

[0191] FIG. **26** illustrates an example of a mirror optic **1704** and transit section **2603**, where the mirror optic comprises only one transit input port **2601***a* and one transit output port **2601***b*. The mirror optic **1704** takes the form of a split aperture mirror featuring two gaps **2602***a*, **2602***b*. A first angled mirror is situated adjacent to the first gap **2302***a*. A second angled mirror is situated adjacent to the second gap **2302***b*. According to other examples, the first and second angled mirrors may be positioned differently with respect to the first and second gaps so long as the mirror **2601***b* is an inverted image of the location of the mirror **2601***a*. The mirror optic **1704** includes one transit input port **2601***a* and one transit output port **2601***b*. As seen in FIG. **26**, the transit input port **2601***a* is located on the first angled mirror and located on one side of the first gap **2302***a*. The transit output port **2601***b* is located on the second angled mirror and located on one side of the second gap **2302***b*. In this case, channels selected for transit at the input transit port **2601***a* are sent directly to the output transit ports **2601***b* via transit section **2603**. There is therefore a direct connection between the input transit port and output transit port without any switching occurring in transit.

[0192] In the arrangement seen in FIG. **26**, the transit section consists of a lens **2604** having equal power along the steering axis as along the dispersion axis. For example, lens **2604** may be a spherical lens. The centre of lens **2604** is aligned with an optical axis which is also aligned with a midpoint between the gaps **2602***a* and **2602***b*, as seen in FIG. **26**. The transit section further includes a flat mirror **2605** positioned normal to the optical axis. The lens **2604** is located at a distance of one focal length from the mirror **2605** and at a distance of one focal length from a midpoint between gap **2602***a* and **2602***b*. The lens **2604** therefore forms an inverted image of the transit input port **2601***a* (in both the steering and dispersion directions) at the transit output port **2601***b*. The second angled mirror then deflects the port to the second programmable deflection plane **1706***b* of the second M×N WSS **1700***b* as previously described. The mirror **2605** seen in FIG. **26** is positioned normal to the optical axis, however according to other examples, the mirror **2605** may be angled with respect to the optical axis such that light incident on the transit section **2603** can be directed to different output ports **1701***b*.

[0193] The demultiplexer **1802** may also comprise fewer or more transit input ports and/or output ports. For example, the demultiplexer **1802** may comprise only one transit input port **2701***a* and one transit output port **2701***b* as seen in FIG. **27**. FIG. **27** illustrates an example of a demultiplexer **1802** and transit section **2703**, where the demultiplexer comprises only one transit input port **2701***a* and one transit output port **2701***b*. Each transit port **2701** is positioned next to a gap in the ports **2102**. Transit input port **2701***a* is positioned next to gap **2702***a* and transit output port **2701***b* is positioned next to gap **2702***b*. The mirrors located at the transit input port **2701***a* and transit output port **2701***b* may be at an angle to the surface of the demultiplexer **1802** but this is not required. The mirrors may be flush with the demultiplexer. In this case, channels selected for transit at the input transit port **2701***a* are sent directly to the output transit ports **2701***b* via transit section **2703**. There is therefore a direct connection between the input transit port and output transit port without any switching.

[0194] The transit section **2703** comprises two lenses **2704***a*, **2704***b* having optical power along the steering axis. The transit section further comprises a lens **2705** having optical power along the dispersion axis. In the example seen in FIG. **27**, lenses **2704**, **2704***b* and **2705** are cylindrical lenses. The transit section further comprises a retroreflecting structure **2706**. In the example seen in FIG. **27**, the retroreflecting structure **2706** comprises two orthogonal mirrors **2707***a*, **2707***b* which are rotated relative to one another about parallel axes A and B. Each mirror is rotated about its respective axis by 45 degrees with respect to the optical axis. The mirrors **2707***a*, **2707***b* are therefore rotated with respect to one another by 90 degrees. According to other examples, the retroreflecting structure may be a prism. The transit section **2703** images the beam incident at transit input port **2701***a* to the mirror at transit output port **2701***b*.

[0195] For the optical path of the chief ray **2707** the lens **2705** has a focal length of a quarter of the distance between the respective mirrors at transit input port **2701***a* and transit output port **2701***b*, thereby forming a 4F imaging system between the two mirrors. The single 4F arrangement of the lens **2705** having optical power in the dispersion direction ensures the correct orientation of the spectra incident on transit output port **2701***b* so that the spectra can be properly multiplexed before exiting the output ports **1701***b*.

[0196] The focal length of each of the lenses **2704***a*, **2704***b* is equal to an eighth of the distance between the respective mirrors at transit input port **2701***a* and transit output port **2701***b*, thereby forming two 4F imaging systems between the two mirrors (a dual 4F arrangement). The dual 4F arrangement formed by lenses **2704***a*, **2704***b* forms a non-inverted real image in the steering direction at transit output port **2701***b* of the spectra incident on transit input port **2701***a*. An inverted image of the spectra incident on the first programmable deflection plane **1706***a* is formed at the second programmable deflection plane **1706***b* meaning there is a one-to-one mapping between the transit input port **2701***a* and the transit output port **2701***b*.

[0197] Although FIGS. **26** and **27** illustrate transit sections **2603** and **2703** being positioned adjacent to the mirror optic **1704** and demultiplexer **1802** respectively, according to further examples, transit section **2603** may be implemented at the demultiplexer and transit section **2703** may be implemented at the mirror optic.

[0198] The edge ROADM switch described herein incorporates two adWSS or two M×N switches. As previously described, each M×N switch **1700***a*, **1700***b* is configured to operate so as to transfer signals from ports **1701***a* to ports **1702***a*, from ports **1701***b* to **1702***b*, from ports **1702***a* to ports **1701***a* and from ports **1702***b* to ports **1701***b*. According to one example, the edge ROADM switch operates so as to transfer signals from ports **1701***a* to ports **1702***a* of M×N switch **1700***a*, from ports **1702***b* to **1701***b* of M×N switch **1700***b*, and, using the transit section as described, from ports **1701***a* to **1701***b* (from M×N switch **1700***a* to switch **1700***b*). It is therefore the case that light from input ports **1701***a* which is not selected by the first programmable deflection plane **1706***a* for transit is deflected to the output ports **1702***a*. Light is which is selected for transit by the first programmable deflection plane **1706***a* and passes through the transit section is therefore merged with light coming from ports **1702***b* at the second programmable deflection plane **1706***b* and together is output at ports **1701***b*. It will be readily understood that further adWSS or M×N switches could be incorporated into a single switching device. Suitably, the programmable deflection planes described herein are SLM planes. The SLM plane may be a MEMS mirror array. The SLM plane may be an LCOS device. The LCoS device applies a hologram to enable a beam deflection. Alternatively, the SLM plane may be another locally configurable device capable of applying a deflection to a channel from the input.

[0199] In the examples described herein, the first and second (and in some cases third and fourth) programmable deflection planes are incorporated onto a single SLM device. This utilises fewer components, reduces control complexity and hence cost compared to if the programmable deflection planes are on separate SLM devices. Alternatively, the programmable deflection planes may be on separate SLM devices.

[0200] In the examples described herein, optical components are used to route data through the described switches. There is no absorption and re-emission of light, thereby avoiding the lag associated with transmitting data through electronical switches. The optical components also have lower power consumption than equivalent electronical implementations.

[0201] The examples described herein incorporate one or more diffraction grating. However, any demultiplexer which demultiplexes light signals into spatially separated data channels may be used instead of a diffraction grating in any of the examples. Similarly, any multiplexer which multiplexes spatially separated data channels into multiplexed light signals may be used instead of a diffraction grating in any of the examples. Any suitable optical dispersion device may be used as a multiplexer and/or a demultiplexer.

[0202] Lenses and other optical components described herein as single structures may be implemented using assemblies having a plurality of components which achieve the same optical effect. Examples of such assemblies are: achromatic doublets, achromatic triplets, Cook doublets, telescopes and microscopes for imaging lenses. The lenses described herein may be implemented using other optics with the same optical power. For example, a curved mirror may be used as a lens. For example, multiple lens elements, mirrors, mirror arrays, catadioptric systems, holographic optical elements or diffractive optical elements may be used as a lens. Separate elements, for example 4F lenses, with the same optical properties can be arranged as separate passes through or from a single physical element.

[0203] The examples described herein generally one or more mirrors. The term "mirror" may be used herein to refer to any reflective surface such as a metallic mirror, an interference layer, a total internal reflection surface (as a prism), a polarisation beamsplitter, or a diffractive or holographic surface.

[0204] The applicant hereby discloses in isolation each individual feature described herein and any combination of two or more such features, to the extent that such features or combinations are capable of being carried out based on the present specification as a whole in the light of the common general knowledge of a person skilled in the art, irrespective of whether such features or combinations of features solve any problems disclosed herein, and without limitation to the scope of the claims. The applicant indicates that aspects of the present invention may consist of any such individual feature or combination of features. In view of the foregoing description it will be evident to a person skilled in the art that various modifications may be made within the scope of the invention.

1. An optical switch comprising:
- a set of first input ports, each first input port configured to transport an optical signal having at least one component frequency channel;
- a set of first output ports, each first output port configured to transport an optical signal having at least one component frequency channel;
- a first programmable deflection plane configured to deflect beams incident on it to form a first deflected array of beams;
- a second programmable deflection plane configured to deflect beams incident on it to form a second deflected array of beams;
- a first demultiplexer configured to separate light from the set of first input ports into its component frequency channels to form the beams incident on the first programmable deflection plane;
- a first multiplexer configured to combine the second deflected array of beams from the second programmable deflection plane into combined signals incident on the set of first output ports; and
- a beam steering optical element group configured to transfer the first deflected array of beams from the first programmable deflection plane to the beams incident on the second programmable deflection plane;
- wherein the incidence normal of the first programmable deflection plane is within the same hemispherical angular area as the incidence normal of the second programmable deflection plane; and
- wherein the dispersion direction in the dispersion plane of spectra of the beams formed on the second programmable deflection plane with respect to the diffraction order of the first multiplexer is such that the combined signals from the first multiplexer match the set of first output ports.

2. An optical switch as claimed in claim **1**, wherein the dispersion direction in the dispersion plane of the first multiplexer is the same sign as the dispersion direction in the dispersion plane of the first demultiplexer.

3. An optical switch as claimed in claim **2**, wherein the first multiplexer is the same as the first demultiplexer.

4. An optical switch as claimed in claim **2**, wherein the beam steering optical element group is configured to form a non-inverted image of the first programmable deflection plane on the second programmable deflection plane in the dispersion plane, and wherein the beam steering optical element group is configured to form a Fourier conjugate image of the first programmable deflection plane on the second programmable deflection plane orthogonal to the dispersion plane.

5. (canceled)

6. An optical switch as claimed in claim **4**, wherein the beam steering optical element group comprises:
- a first optical element and a second optical element, the first and second optical elements configured to divert the first deflected array of beams from the first programmable deflection plane towards a beam steering optical device;
- the beam steering optical device configured to redirect the diverted first deflected array of beams to form a redirected first deflected array of beams; and
- a third optical element and a fourth optical element, the third and fourth optical elements configured to divert the redirected first deflected array of beams from the beam steering optical device towards the second programmable deflection plane.

7. An optical switch as claimed in claim **6**, wherein each of the first optical element, second optical element, third optical element, fourth optical element and beam steering optical device has optical power.

8. An optical switch as claimed in claim **1**, wherein the dispersion direction in the dispersion plane of the first multiplexer is the opposite sign as the dispersion direction in the dispersion plane of the first demultiplexer.

9. An optical switch as claimed in claim **8**, wherein the beam steering optical element group is configured to form an inverted image of the first programmable deflection plane on the second programmable deflection plane in the dispersion plane, and wherein the beam steering optical element group is configured to form a Fourier conjugate image of the first programmable deflection plane on the second programmable deflection plane orthogonal to the dispersion plane.

10. (canceled)

11. An optical switch as claimed in claim **9**, wherein the beam steering optical element group comprises:
- a first optical element configured to divert the first deflected array of beams from the first programmable deflection plane towards a beam steering optical device;
- the beam steering optical device configured to redirect the diverted first deflected array of beams to form a redirected first deflected array of beams; and
- a second optical element configured to divert the redirected first deflected array of beams from the beam steering optical device towards the second programmable deflection plane.

12. An optical switch as claimed in claim **11**, wherein each of the first optical element, second optical element and beam steering optical device has optical power.

13. An optical switch as claimed in claim **1**, wherein the beam steering optical element group comprises a first lens assembly, wherein the first lens assembly comprises a spherical lens and a cylindrical lens with power only in the dispersion axis of the cylindrical lens.

14. (canceled)

15. An optical switch as claimed in claim **13**, further comprising:

a first mirror segment configured to divert a portion of the first deflected array of beams from the first programmable deflection plane to a curved mirror; and

a second mirror segment configured to divert the portion of the first deflected array of beams from the curved mirror to the second programmable deflection plane.

16. An optical switch as claimed in claim **13**, further comprising:

a first mirror segment, a second mirror segment and a further beam steering optical device;

the first mirror segment configured to divert a portion of the first deflected array of beams from the first programmable deflection plane to the further beam steering optical device;

the further beam steering optical device configured to remap the portion of the first deflected array of beams to form a remapped portion of beams directed to the second mirror segment; and

the second mirror segment configured to divert the remapped portion of beams from the further beam steering optical device to the curved mirror, and from the curved mirror to the second programmable deflection plane.

17. A reconfigurable optical add-drop multiplexer (ROADM) optical switch comprising the optical switch of claim **1**, and further comprising:

a set of second input ports, each second input port configured to transport an optical signal having at least one component frequency channel;

a set of second output ports, each second output port configured to transport an optical signal having at least one component frequency channel;

a third programmable deflection plane configured to deflect beams from the set of second input ports incident on it to form a third deflected array of beams;

a fourth programmable deflection plane configured to deflect beams incident on it to form a fourth deflected array of beams towards the set of second output ports;

a second demultiplexer configured to separate light from the third programmable deflection plane into its component frequency channels towards the second programmable deflection plane;

a second multiplexer configured to combine the first deflected array of beams from the first programmable deflection plane into combined signals towards the fourth programmable deflection plane;

wherein the incidence normal of each of the first, second, third and fourth programmable deflection planes is within the same hemispherical angular area; and

wherein the first demultiplexer and second multiplexer have the same dispersion direction; and

wherein the second demultiplexer and first multiplexer have the same dispersion direction.

18. A reconfigurable optical add-drop multiplexer (ROADM) optical switch as claimed in claim **17**, wherein the first demultiplexer is the same as the second multiplexer, wherein the second demultiplexer is the same as the first multiplexer.

19. (canceled)

20. A reconfigurable optical add-drop multiplexer (ROADM) optical switch as claimed in claim **17**, comprising a single spatial light modulator (SLM) panel comprising the first, second, third and fourth programmable deflection planes.

21. An optical switch as claimed in claim **1**, comprising a single spatial light modulator (SLM) panel comprising the first and second programmable deflection planes.

22. An optical switch as claimed in claim **1**, wherein the switch further comprises a mirror optic, the mirror optic comprising a pair of mirrors separated by a gap; and a further mirror optic, the further mirror optic comprising a pair of mirrors separated by a gap.

23-25. (canceled)

26. An optical switch as claimed in claim **13**, wherein the beam steering optical element group comprises a second lens assembly, and wherein the second lens assembly comprises a spherical lens and a cylindrical lens with power only in the dispersion axis of the cylindrical lens.

27. (canceled)

28. An optical switch as claimed in claim **26**, wherein the first lens assembly comprises the first demultiplexer and the second lens assembly comprises the first multiplexer.

29-41. (canceled)

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